

EXHIBIT 47

8672626



THE UNITED STATES OF AMERICA

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UNITED STATES DEPARTMENT OF COMMERCE

United States Patent and Trademark Office

January 16, 2026

THIS IS TO CERTIFY THAT ANNEXED IS A TRUE COPY FROM THE
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OF:

APPLICATION NUMBER: *13/096,511*

FILING DATE: *April 28, 2011*

PATENT NUMBER: *8330235*

ISSUE DATE: *December 11, 2012*



Certified by

DIRECTOR OF THE UNITED STATES PATENT AND TRADEMARK OFFICE

GF-ITC-00004310

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:	Karthik RAMANI <i>et al.</i>	Confirmation No.:	Not yet assigned
Application No.:	Not yet assigned	Examiner:	Not yet assigned
Filed:	Herewith	Group Art Unit:	Not yet assigned

For: METHOD TO REDUCE MOL DAMAGE ON NiSi

Commissioner for Patents
Alexandria, VA 22313-1450

PRELIMINARY AMENDMENT

Dear Sir:

Prior to consideration of the above-identified application, please amend the application based on the following.

AMENDMENT AND PRESENTATION OF CLAIMS 2

REMARKS 7

AMENDMENT AND PRESENTATION OF CLAIMS

Please replace all prior claims in the present application with the following claims.

1. - 13. (Canceled)

14. (Original) A semiconductor device comprising:

a silicon substrate,

a transistor, comprising source/drain regions and a gate electrode, on the substrate; and

a nickel silicide layer on the gate electrode and source/drain regions,

wherein the nickel silicide layer comprises platinum having a composition gradient with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.

15. (Canceled)

16. (Original) The semiconductor device according to claim 14, wherein the concentration gradient of platinum in the nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

17. (Original) The semiconductor device according to claim 16, wherein the concentration gradient of platinum in the nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

18. (Original) The semiconductor device according to claim 14, wherein the nickel silicide layer has a thickness of about 100 Å to about 220 Å.

19. (Original) The semiconductor device according to claim 14, wherein the nickel silicide layer has a thickness of about 100 Å to about 150 Å.

20. (Canceled)

21. (New) The semiconductor device according to claim 14, wherein the silicon substrate comprises a strained silicon germanium (Si-Ge) layer on a silicon layer.

22. (New) The semiconductor device according to claim 14, further comprising a stress liner on the nickel silicide layer.

23. (New) The semiconductor device according to claim 22, further comprising an oxide or oxynitride layer on the stress liner.

24. (New) The semiconductor device according to claim 23, wherein the stress liner comprises stressed silicon nitride.

25. (New) A semiconductor device comprising:

a silicon substrate,

a P-channel transistor and an N-channel transistor, each comprising source/drain regions and

a gate electrode, on the silicon substrate; and

a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor; and

a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor,

wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient with platinum increasing in amount in a direction away from the P-channel and N-channel transistors.

26. (New) The semiconductor device according to claim 25, further comprising:

a first stress liner on the P-channel transistor; and

a second stress liner on the N-channel transistor.

27. (New) The semiconductor device according to claim 26, wherein the first stress liner exhibits high compressive stress and the second stress liner exhibits high tensile stress.

28. (New) The semiconductor device according to claim 27, wherein the concentration gradient of platinum in each nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

29. (New) The semiconductor device according to claim 28, wherein the concentration gradient of platinum in each nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

30. (New) The semiconductor device according to claim 27, wherein each nickel silicide layer has a thickness of about 100 Å to about 220 Å.

31. (New) The semiconductor device according to claim 30, wherein each nickel silicide layer has a thickness of about 100 Å to about 150 Å.

32. (New) The semiconductor device according to claim 27, wherein the silicon substrate comprises a strained silicon germanium (Si-Ge) layer on a silicon layer.

33. (New) The semiconductor device according to claim 27, further comprising an oxide or oxynitride layer on the first and second stress liners.

34. (New) The semiconductor device according to claim 33, wherein the first and second stress liners comprise stressed silicon nitride.

35. (New) A semiconductor device comprising:

a silicon substrate;

a strained silicon germanium (Si-Ge) layer on the silicon substrate;

a P-channel transistor comprising source/drain regions and a gate electrode, on the strained Si-Ge layer;

an N-channel transistor comprising source/drain regions and a gate electrode, on the strained Si-Ge layer, spaced from the P-channel transistor;

a first nickel silicide layer, having a thickness of about 100 Å to about 220 Å, on the gate electrode and source/drain regions of the P-channel transistor;

a second nickel silicide layer, having a thickness of about 100 Å to about 220 Å, on the gate electrode and source/drain regions of the N-channel transistor;

a compressively stressed silicon nitride layer on the first nickel silicide layer;

a tensile stressed silicon nitride layer on the second nickel silicide layer;

a first oxide or oxynitride layer on the compressively stressed silicon nitride layer; and

a second oxide or oxynitride layer on the tensile stressed silicon nitride layer,

wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient with platinum increasing in amount from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. % in a direction away from the P-channel and N-channel transistors.

REMARKS

By this amendment, claims 14, 16 through 19, and 21 through 35 are pending, in which claims 1 through 13, 15, and 20 are canceled without prejudice or disclaimer, no claims are withdrawn from consideration or currently amended, and claims 21 through 35 are newly presented. No new matter is introduced.

Previously allowed method claims 1 through 13 and 20 have been canceled. In addition prior claim 15 has been canceled and rewritten as independent claim 25.

Favorable consideration is respectfully requested. If any unresolved issues remain, it is respectfully requested that the Examiner telephone the undersigned attorney at (703) 519-9954 so that such issues may be resolved as expeditiously as possible.

To the extent necessary, a petition for an extension of time under 37 C.F.R. § 1.136 is hereby made. Please charge any shortage in fees due in connection with the filing of this paper, including extension of time fees, to Deposit Account 504213 and please credit any excess fees to such deposit account.

Respectfully Submitted,

DITTHAVONG MORI & STEINER, P.C.

April 28, 2011
Date

/Arthur J. Steiner/
Arthur J. Steiner
Attorney/Agent for Applicant(s)
Reg. No. 26106

Anita Pellman Gross
Attorney/Agent for Applicant(s)
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Tel. (703) 519-9951
Fax (703) 519-9958

COMBINED DECLARATION AND POWER OF ATTORNEY
FOR PATENT APPLICATION

As a below named inventor, I hereby declare that:

My residence, post office address and citizenship are as stated below, next to my name.

I believe I am the original, first, and joint inventor of the subject matter which is claimed and for which a patent is sought on the invention entitled:

METHOD TO REDUCE MOL DAMAGE ON NISI

the specification of which is attached hereto.

I hereby state that I have reviewed and understand the contents of the above-identified specification, including the claim(s), as amended by any amendment referred to above.

I acknowledge the duty to disclose all information known to me to be material to patentability as defined in Title 37, Code of Federal Regulations, Section 1.56 (copy attached).

I hereby claim foreign priority benefits under Title 35, United States Code, Section 119(a)-(d), on any foreign application(s) for patent or inventor's certificate listed below and have also identified below any foreign application for patent or inventor's certificate having a filing date before that of the application on which priority is claimed:

Prior Foreign Application(s)		Priority Claimed		
(Number)	(Country)	(Day/Month/Year Filed)	Yes	No

I hereby claim the benefit under Title 35, United States Code, Section 119(e) of any United States provisional application(s) listed below:

I hereby claim benefit under Title 35, United States Code, Section 120 of any United States application(s) listed below and, insofar as the subject matter of each of the claims of this application is not disclosed in the prior United States application in the manner provided by the first paragraph of Title 35, United States Code, Section 112, I acknowledge the duty to disclose all information known to me to be material to patentability as defined in Title 37, Code of Federal Regulations, Section 1.56 (copy attached) which became available between the filing date of the prior application and the national or PCT International filing date of this application:

As named inventor, I hereby revoke all powers of attorney previously given and appoint the practitioners associated with the following Customer No.:

Customer Number:

30671

with full power of substitution and revocation, to prosecute this application and to transact all business in the Patent and Trademark Office connected therewith.

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

Send all future correspondence to the attention of Arthur J. Steiner, Reg. No. 26106, care of the address associated with the following customer number, and direct all telephone calls to the same at (703) 519-9954.

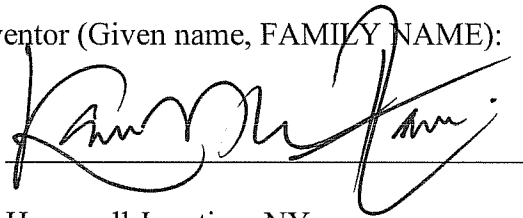
Customer Number:

30671

Full Name of First Inventor (Given name, FAMILY NAME):

Karthik RAMANI

Inventor's Signature



Date

29 Jan 2009

Residence:

Hopewell Junction, NY

Citizenship India

Post Office Address

3603 Chelsea Cove South, Hopewell Junction, NY 12533

Full Name of Second Inventor (given name, FAMILY NAME):

Paul R. BESSER

Inventor's Signature

Date

Residence:

Sunnyvale, CA

Citizenship US

Post Office Address

1087 Yorktown Drive, Sunnyvale, CA 94087

Title 37, Code of Federal Regulations, Section 1.56
Duty to Disclose Information Material to Patentability

(a) A patent by its very nature is affected with a public interest. The public interest is best served, and the most effective patent examination occurs when, at the time an application is being examined, the Office is aware of and evaluates the teachings of all information material to patentability. Each individual associated with the filing and prosecution of a patent application has a duty of candor and good faith in dealing with the Office, which includes a duty to disclose to the Office all information known to that individual to be material to patentability as defined in this section. The duty to disclose information exists with respect to each pending claim until the claim is canceled or withdrawn from consideration, or the application becomes abandoned. Information material to the patentability of a claim that is canceled or withdrawn from consideration need not be submitted if the information is not material to the patentability of any claim remaining under consideration in the application. There is no duty to submit information which is not material to the patentability of any existing claim. The duty to disclose all information known to be material to patentability is deemed to be satisfied if all information known to be material to patentability of any claim issued in a patent was cited by the Office or submitted to the Office in the manner prescribed by §§ 1.97(b)-(d) and 1.98. However, no patent will be granted on an application in connection with which fraud on the Office was practiced or attempted or the duty of disclosure was violated through bad faith or intentional misconduct. The Office encourages applicants to carefully examine:

- (1) Prior art cited in search reports of a foreign patent office in a counterpart application, and
- (2) The closest information over which individuals associated with the filing or prosecution of a patent application believe any pending claim patentably defines, to make sure that any material information contained therein is disclosed to the Office.

(b) Under this section, information is material to patentability when it is not cumulative to information already of record or being made of record in the application, and

- (1) It establishes, by itself or in combination with other information, a prima facie case of unpatentability of a claim; or
- (2) It refutes, or is inconsistent with, a position the applicant takes in:
 - (i) Opposing an argument of unpatentability relied on by the Office, or
 - (ii) Asserting an argument of patentability.

A prima facie case of unpatentability is established when the information compels a conclusion that a claim is unpatentable under the preponderance of evidence, burden-of-proof standard, giving each term in the claim its broadest reasonable construction consistent with the specification, and before any consideration is given to evidence which may be submitted in an attempt to establish a contrary conclusion of patentability.

(c) Individuals associated with the filing or prosecution of a patent application within the meaning of this section are:

- (1) Each inventor named in the application;
- (2) Each attorney or agent who prepares or prosecutes the application; and
- (3) Every other person who is substantively involved in the preparation or prosecution of the application and who is associated with the inventor, with the assignee or with anyone to whom there is an obligation to assign the application.

(d) Individuals other than the attorney, agent or inventor may comply with this section by disclosing information to the attorney, agent, or inventor.

COMBINED DECLARATION AND POWER OF ATTORNEY
FOR PATENT APPLICATION

As a below named inventor, I hereby declare that:

My residence, post office address and citizenship are as stated below, next to my name.

I believe I am the original, first, and joint inventor of the subject matter which is claimed and for which a patent is sought on the invention entitled:

METHOD TO REDUCE MOL DAMAGE ON NISI

the specification of which is attached hereto.

I hereby state that I have reviewed and understand the contents of the above-identified specification, including the claim(s), as amended by any amendment referred to above.

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Customer Number:

30671

Full Name of First Inventor (Given name, FAMILY NAME): Karthik RAMANI

Inventor's Signature _____ Date _____

Residence: Hopewell Junction, NY Citizenship India

Post Office Address 3603 Chelsea Cove South, Hopewell Junction, NY 12533

Full Name of Second Inventor (given name, FAMILY NAME): Paul R. BESSER

Inventor's Signature Paul R Besser Date 1/29/09

Residence: Sunnyvale, CA Citizenship US

Post Office Address 1087 Yorktown Drive, Sunnyvale, CA 94087

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- (1) Prior art cited in search reports of a foreign patent office in a counterpart application, and
- (2) The closest information over which individuals associated with the filing or prosecution of a patent application believe any pending claim patentably defines, to make sure that any material information contained therein is disclosed to the Office.

(b) Under this section, information is material to patentability when it is not cumulative to information already of record or being made of record in the application, and

- (1) It establishes, by itself or in combination with other information, a prima facie case of unpatentability of a claim; or
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(c) Individuals associated with the filing or prosecution of a patent application within the meaning of this section are:

- (1) Each inventor named in the application;
- (2) Each attorney or agent who prepares or prosecutes the application; and
- (3) Every other person who is substantively involved in the preparation or prosecution of the application and who is associated with the inventor, with the assignee or with anyone to whom there is an obligation to assign the application.

(d) Individuals other than the attorney, agent or inventor may comply with this section by disclosing information to the attorney, agent, or inventor.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:	Karthik RAMANI <i>et al.</i>	Confirmation No.:	Not yet assigned
Application No.:	Not yet assigned	Examiner:	Not yet assigned
Filed:	Herewith	Group Art Unit:	Not yet assigned

For: METHOD TO REDUCE MOL DAMAGE ON NiSi

Commissioner for Patents
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Sir:

This Information Disclosure Statement is submitted:

- ☒ Under 37 CFR § 1.97(b): (1) Within three months of filing of national application; (2) Within three months of date of entry of the national stage as set forth in § 1.491 in an international application; (3) Before mailing of a first Office action on the merits; or (4) Before mailing of a first Office action after the filing of a request for continued examination under § 1.114.
- ☐ Under 37 CFR § 1.97(c): After the period specified in § 1.97(b) but before the mailing date of either a final action under § 1.113 or a notice of allowance under § 1.311, whichever occurs first, and accompanied by either:
- ☐ (1) A statement under § 1.97(e) as set forth below, or
- ☐ (2) A \$180.00 fee under 37 CFR 1.17(p).
- ☐ Under 37 CFR § 1.97(d): After the period specified in § 1.97(c) but on or before payment of the issue fee, and accompanied by:
- ☐ (1) A statement under § 1.97(e) as set forth below; and
- ☐ (2) A \$180.00 fee set forth in 37 CFR 1.17(p).

Statement under 37 CFR § 1.97(e):

- ☒ Not necessary, because this IDS is filed under § 1.97(b) or § 1.97(c)(2).
- ☐ Each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a

counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1); or

- ☐ No item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

Form PTO 1449–Information Disclosure Citation is attached with copies of patents, publications, or other information of which Applicants are aware, which may be material to the examination of this application, and for which there may be a duty to disclose in accordance with 37 CFR 1.56. This is not to be construed as a representation that a search has been made or that no better prior art exists, or that a reference is relevant merely because cited.

The Examiner is requested to initial the attached form PTO-1449 and to return a copy of the initialed document to the undersigned as an indication that the attached references have been considered and made of record.

To the extent necessary, a petition for an extension of time under 37 C.F.R. § 1.136 is hereby made. Please charge any shortage in fees due in connection with the filing of this paper, including extension of time fees, to Deposit Account 504213 and please credit any excess fees to such deposit account.

Respectfully submitted,

DITTHAVONG MORI & STEINER, P.C.

/Arthur J. Steiner/

Arthur J. Steiner

Attorney/Agent for Applicant(s)

Reg. No. 26106

918 Prince Street
Alexandria, VA 22314
Tel: (703) 519-9954
Fax: (703) 519-9958

Date: April 28, 2011

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:	Karthik RAMANI <i>et al.</i>	Confirmation No.:	Not yet assigned
Application No.:	Not yet assigned	Examiner:	Not yet assigned
Filed:	Herewith	Group Art Unit:	Not yet assigned

For: METHOD TO REDUCE MOL DAMAGE ON NiSi

Commissioner for Patents
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

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- ☐ (1) A statement under § 1.97(e) as set forth below, or
- ☐ (2) A \$180.00 fee under 37 CFR 1.17(p).
- ☐ Under 37 CFR § 1.97(d): After the period specified in § 1.97(c) but on or before payment of the issue fee, and accompanied by:
- ☐ (1) A statement under § 1.97(e) as set forth below; and
- ☐ (2) A \$180.00 fee set forth in 37 CFR 1.17(p).

Statement under 37 CFR § 1.97(e):

- ☒ Not necessary, because this IDS is filed under § 1.97(b) or § 1.97(c)(2).
- ☐ Each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a

counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1); or

- ☐ No item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

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Respectfully submitted,

DITTHAVONG MORI & STEINER, P.C.

/Arthur J. Steiner/

Arthur J. Steiner

Attorney/Agent for Applicant(s)

Reg. No. 26106

918 Prince Street
Alexandria, VA 22314
Tel: (703) 519-9954
Fax: (703) 519-9958

Date: April 28, 2011

Under the Paperwork reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449A/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	Not yet assigned
				Filing Date	04-28-11
				First Named Inventor	Karthik Ramani
				Art Unit	Not yet assigned
				Examiner Name	Not yet assigned
				Attorney Docket Number	P2343US01
Sheet	1	of	1		

[illegible][illegible]

Examiner Signature		Date Considered	
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* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. 1. Applicant's unique citation design number (optional). 2 See Kinds Codes of USPTO patent Documents. at www.uspto.gov or MPEP 901.04. 3. Enter Office that issued the document, by the two-letter code (WIPO Standard ST.3). 4. For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. 5. Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST. 16 if possible. 6. Applicant is to place a check mark here if English language Translation is attached.

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 2 hours to complete, including gathering, preparing, and submitting the completed application form the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, P.O. Box 1450 Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS.

SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 (1-800-786.9199) and select option 2.

METHOD TO REDUCE MOL DAMAGE ON NiSi

RELATED APPLICATIONS

[0001] This application is a Divisional Application of U.S. Application Serial No. 12/366,378 filed on February 5, 2009.

TECHNICAL FIELD

[0002] The present disclosure relates to semiconductor devices with improved nickel silicide layers. The present disclosure is particularly applicable in fabricating transistors with improved device performance by avoiding nickel silicide degradation.

BACKGROUND

[0003] Conventional semiconductor devices, such as the one shown in FIG. 1, include a refractory metal deposited over all exposed upper surface features of a silicon substrate 101, preferably by means of physical vapor deposition (PVD). The most common refractory metal employed is nickel, which forms a very low resistivity silicide with silicon and does not consume as much Si as other silicides commonly employed. NiSi has a great resistance to etchants which are used to etch silicon oxide, a common insulating material used in semiconductor devices.

[0004] The refractory metal can be sputtered by DC magnetron sputtering in an ultra-high vacuum, multi-chamber system. The formation of an MOS transistor requires the formation of a gate structure 103 and source/drain junctions 105. The gate electrode 103 typically is formed by depositing a layer of heavily doped polysilicon 103a on a metal oxide insulating layer 103b and etching the layers to pattern the electrode. Sidewall spacers 107 are deposited on the opposing surfaces of the patterned gate electrode with oxide liner 109 between spacer 107 and both gate electrode 103 and substrate 101. The sidewall spacers 107 comprise silicon oxide, silicon nitride or silicon oxynitride.

[0005] Generally, in forming the MOS transistor, the refractory metal is deposited after etching the gate electrode 103 and after forming the source/drain junction 105, as illustrated in FIG. 1. After deposition, the refractory metal layer blankets the top surface of the gate electrode 103, sidewall spacers 107, oxide liners 109, and source/drain regions 105. As a result of thermal processing, e.g., a rapid thermal annealing (RTA) process, performed in an inert or reducing atmosphere, the refractory metal reacts with underlying silicon to form electrically conductive silicide layer portions 111 on the top surface of the polysilicon gate electrode 103 and on the exposed surfaces of the substrate where source and drain regions 105 are formed. Unreacted portions of the refractory metal layer, e.g., on the silicon oxide, nitride or oxynitride sidewall spacers 107 and the silicon oxide liner 109 are then removed, as by a wet etching process selective to the metal silicide portions 111.

[0006] Substrates based on “strained silicon” have attracted interest as a semiconductor material which provides increased speeds of electron and hole flow therethrough, thereby permitting fabrication of semiconductor devices with higher operating speeds, enhanced performance characteristics, and lower power consumption. A layer 101a of silicon-germanium (Si-Ge) is formed on a suitable crystalline substrate, e.g., a Si wafer or a silicon-on-insulator (SOI) wafer 101b. The Si-Ge layer has a greater lattice constant (spacing of Si and Ge atoms therein) relative to the underlying Si. As a consequence, the Si-Ge layer has a compressive strain.. Electrons in such compressively strained layers have greater mobility than in conventional Si layers with smaller inter-atom spacings, i.e., there is less resistance to electron flow. For example, electron flow may be up to about 70% faster compared to electron flow in conventional Si. Transistors and IC devices formed with such strained Si layers can exhibit operating speeds up to about 35% faster than those of equivalent devices formed with conventional Si, without necessity for reduction in transistor size. Conventional practices based on strained silicon technology also involve epitaxially growing a relaxed silicon layer on a tensilely stressed silicon layer which is subsequently doped to form relaxed source/drain regions in the relaxed silicon layer.

[0007] Conventional semiconductor devices typically comprise a plurality of active devices in or on a common semiconductor substrate, e.g., CMOS devices comprising at least a pair of PMOS and NMOS transistors in spaced adjacency. The mobility of electrons is faster than the mobility

of holes in conventional bulk silicon substrates. Accordingly, in conventional CMOS transistors, the drive current of the PMOS transistor is less than the drive current of the NMOS transistor creating an imbalance. This imbalance is exacerbated in CMOS transistors fabricated on or within a tensilely stressed active device area formed in a strained lattice semiconductor substrate, e.g., strained Si-Ge on Si, because the increase in electron mobility is greater than the increase in hole mobility.

[0008] Channel carrier mobility of transistors based on strained Si substrates can be increased by applying a stress thereto. In forming P-channel transistors, channel carrier mobility is enhanced by applying a stressed dielectric layer exhibiting high compressive stress for increasing electron mobility. In N-channel transistors, channel carrier mobility is significantly increased by applying a stressed layer exhibiting high tensile stress for increasing hole mobility. Stressed dielectric layers, called stress liners, are applied to transistors wherein the source/drain regions are formed within the strained Si layer, and to transistors having relaxed source/drain regions formed on strained Si layers. The stress liner may comprise silicon carbide, silicon nitride or silicon oxynitride, and may be deposited by plasma enhanced chemical vapor deposition (PECVD) at a thickness of about 200 Å to about 1000 Å. Conventional PECVD conditions may be employed for deposition of a highly compressive layer or highly tensile dielectric layer. In depositing a stress liner exhibiting high compressive stress, both high frequency and low frequency power are applied. When depositing a stressed dielectric layer exhibiting high tensile stress, the low frequency power is significantly reduced.

[0009] Because the stress liners differ for P-channel and N-channel transistors, the stressed dielectric layers are typically patterned by reactive ion etching, which involves physically bombarding the surfaces to subtractively remove material. This etching process involves an “overetch” in order to clear the residue left behind. It is during this overetch process that the underlying NiSi layers are damaged, causing degradation of the NiSi. The degradation can be morphological, wherein mouse bites or missing NiSi is observed, as illustrated in FIG. 1 at 113a-113f, or electrical, wherein the resistivity is increased. Either or both types of degradation ultimately result in poor device parameters such as higher sheet resistance and external resistance.

[0010] A need therefore exists for methodology enabling fabrication of semiconductor devices comprising transistors with degradation resistant NiSi layers and for the resulting improved semiconductor devices having reduced resistance and improved performance.

SUMMARY

[0011] An aspect of the present disclosure is a semiconductor device comprising a transistor with reduced resistance.

[0012] Another aspect of the present disclosure is an efficient method of fabricating a semiconductor device comprising a transistor with reduced resistance.

[0013] Additional aspects and other features of the present disclosure will be set forth in the description which follows and in part will be apparent to those having ordinary skill in the art upon examination of the following or may be learned from the practice of the present disclosure. The advantages of the present disclosure may be realized and obtained as particularly pointed out in the appended claims.

[0014] According to the present disclosure, some technical effects may be achieved in part by a method comprising: forming a transistor on a substrate, the transistor including source/drain regions and a gate electrode; forming a first layer of nickel containing a first amount of platinum over the transistor; forming a second layer of nickel containing a second amount of platinum on the first layer of nickel, the second amount of platinum being greater than the first amount of platinum; and annealing the first and second layers of nickel to form a layer of NiSi over the source/drain regions and the gate electrode, wherein the NiSi layer has a platinum composition gradient with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.

[0015] Aspects of the present disclosure include forming the first and second layers of nickel at a combined thickness of about 45 Å to about 100 Å with the first layer of nickel at a thickness of about 60 % to about 80 % of the combined thickness. Aspects also include annealing to form the layer of NiSi at a thickness of about 100 Å to about 220 Å. Further aspects include annealing by heating at a temperature of about 340° C to about 450°C, such as at a temperature of about 360°

C to about 400° C. This technique can be used to effect annealing in a single step. Another aspect includes annealing by heating at a first stage at a first temperature of about 280° C to about 350° C, for example about 300° C to about 320° C, followed by heating at a second stage at a second temperature of about 360° C to about 450°C, for example about 380° C to about 420° C. Additional aspects include forming the first layer of nickel with a platinum content of about 8 weight % (“wt. %”) to about 12 wt. %, for example about 10 wt. %, and the second layer of nickel with a platinum content of about 30 wt. % to about 70 wt. %, for example about 30 wt. % to about 50 wt. %. Aspects also include forming a capping layer of titanium nitride over the second layer before annealing, and removing the capping layer after annealing, followed by etching to remove unreacted nickel.

[0016] Another aspect of the present disclosure is a semiconductor device comprising: a silicon substrate; a transistor, comprising source/drain regions and a gate electrode, on the substrate; and a NiSi layer on the gate electrode and source/drain regions, the NiSi layer comprising platinum having a composition gradient, the platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.

[0017] Aspects include devices with a P-channel transistor, an N-channel transistor, a first NiSi layer on the P-channel transistor, a second NiSi layer on the N-channel transistor, the first and second NiSi layers each comprising platinum having a concentration gradient with platinum increasing in amount toward the upper surface, i.e., increasing in amount in a direction away from the P-channel and N-channel transistors, a first stress liner on the P-channel transistor, and a second stress liner on the N-channel transistor. Further aspects include the NiSi layers having a thickness of about 100 Å to about 220 Å, for example about 100 Å to about 150 Å.

[0018] Another aspect of the present disclosure is a method of fabricating a semiconductor device, the method comprising: forming a P-channel and an N-channel transistor on a silicon substrate, each transistor including a polysilicon gate electrode and source/drain regions formed in the silicon substrate; depositing a first layer of nickel containing a first amount of platinum on both transistors; depositing a second layer of nickel having a second amount of platinum on the first layer of nickel, the second amount of platinum being greater than the first amount of platinum; depositing a titanium nitride capping layer on the second layer of nickel, annealing the

first and second layers of nickel to form a single layer of NiSi containing platinum having a concentration gradient with the platinum increasing in amount in a direction away from the gate electrode and source/drain regions; etching to remove the capping layer and unreacted nickel; forming a first stress liner exhibiting high compressive stress over both transistors; selectively removing the first stress liner from the N-channel transistor; forming a second stress liner exhibiting high tensile stress over both transistors; and selectively removing the second stress liner from the P-channel transistor.

[0019] Additional aspects and technical effects of the present disclosure will become readily apparent to those skilled in the art from the following detailed description wherein embodiments of the present disclosure are described simply by way of illustration of the best mode contemplated to carry out the present disclosure. As will be realized, the present disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects, all without departing from the present disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] The present disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawing and in which like reference numerals refer to similar elements and in which:

[0021] FIG. 1 schematically illustrates a conventional transistor device with a degraded NiSi layer;

[0022] FIGS. 2 through 6 schematically illustrate sequential steps of a method in accordance with an embodiment of the present disclosure;

[0023] FIGS. 7 through 15 schematically illustrate sequential steps of a method in accordance with another embodiment of the present disclosure.

DETAILED DESCRIPTION

[0024] In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of exemplary embodiments. It should be apparent, however, that exemplary embodiments may be practiced without these specific details or with an equivalent arrangement. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring exemplary embodiments.

[0025] The present disclosure addresses and solves the problem of NiSi layer degradation during subsequent processing, as during middle of the line processing to remove overlying stress liners, which NiSi layer degradation causes resistance problems and adversely affects transistor performance. In accordance with embodiments of the present disclosure, a transistor device is provided having a NiSi layer formulated to resist degradation, advantageously using methodology that does not impact process flow. In accordance with embodiments of the present disclosure, NiSi layer damage is prevented or significantly reduced by formulating the NiSi layer with a platinum composition gradient with the amount of platinum increasing in a direction toward its upper surface, i.e., in a direction away from the source/drain regions and the gate electrode, thereby shielding the NiSi layer from physical damage during post NiSi processing as when removing an overlying stress liner by etching.

[0026] Embodiments of the present disclosure include forming two layers of nickel on a silicon substrate and a silicon gate electrode, the first layer having a relatively low concentration of platinum and the second having a relatively high concentration of platinum, and annealing the two layers to form a single layer of NiSi. During annealing, platinum diffuses to form a composition gradient with the amount of platinum increasing in a direction away from the substrate and gate electrode. It was found that such a concentration gradient of platinum effectively shields the NiSi layer from physical damage during subsequent processing, e.g., etching, while concomitantly maintaining the integrity of the NiSi layer.

[0027] Methodology in accordance with embodiments of the present disclosure includes forming forming a transistor on a substrate, the transistor including source/drain regions and a gate electrode, depositing a first layer of nickel containing a relatively low amount of platinum, for

example about 8 wt. % to about 12 wt. %, e.g., about 10 wt. %, over the transistor, forming a second layer of nickel containing a relatively high amount of platinum, such as about 30 wt. % to about 70 wt. %, e.g., about 30 wt. % to about 50 wt. %, on the first layer. The first and second layers of nickel are formed at a combined thickness of about 45 Å to about 100 Å, with the first layer of nickel at a thickness of about 60 % to about 80 % of the combined thickness. A capping layer of silicon nitride is formed over the second nickel layer. An RTA process is then implemented to convert the platinum containing nickel layers into a single layer of NiSi over the source/drain regions and the gate electrode, such that the NiSi layer has a platinum composition gradient with the amount of platinum increasing in a direction away from the source/drain regions and the gate electrode. The resulting platinum concentration gradient protects the NiSi film without increasing the resistivity of the layer.

[0028] The annealing process may be controlled to form a layer of NiSi at a thickness of about 100 Å to about 220 Å. Annealing may comprise a single heating stage at a temperature of about 340° C to about 450° C, such as at a temperature of about 360° C to about 400° C. Alternatively, a two stage annealing process can be performed, as by heating at a first temperature of about 280° C to about 350° C, such as about 300° C to about 320° C, followed by heating at a second temperature of about 360° C to about 450° C, such as about 380° C to about 420° C. In accordance with embodiments of the present disclosure, the capping layer is removed after annealing followed by etching to remove unreacted nickel, employing conventional methodology.

[0029] Embodiments of the present disclosure include forming two transistors, one P-channel and one N-channel transistor. A first stress liner exhibiting high compressive stress can be formed over both transistors, and the first stress liner is selectively removed, as by etching, from the N-channel transistor. A second stress liner exhibiting high tensile stress can be subsequently formed over both transistors, and the second stress liner is selectively removed, as by etching, from the P-channel transistor. During etching of the stress liners, the thus formed platinum concentration gradient prevents the NiSi from being etched and damaged.

[0030] Still other aspects, features, and technical effects will be readily apparent to those skilled in this art from the following detailed description, wherein preferred embodiments are shown and

described, simply by way of illustration of the best mode contemplated. The disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

[0031] A method in accordance with an embodiment of the present disclosure is schematically illustrated in FIGS. 2 through 5, wherein like reference characters denote like features. Adverting to FIG. 2, strained Ge-Si layer 201 is formed over Si layer 203. Ge-Si layer 201 may be globally strained or locally strained under the source/drain regions. Gate electrode 205 is formed over strained Ge-Si layer 201 with a gate dielectric 207 therebetween. Lightly doped source/drain regions 213a are formed in Ge-Si layer 201 using the gate electrode 205 as a mask in a conventional manner. A sidewall spacer comprising a silicon oxide liner 209, as at a thickness of about 60 Å to about 600 Å, is formed on side surfaces of the gate electrode 205 and on a portion of the upper surface of strained Ge-Si layer 201. Silicon nitride spacers 211 are then formed on the silicon oxide liner 209 as by employing PECVD followed by etching.

[0032] It should be appreciated that liner 209 may be deposited by ALD and may also comprise silicon nitride. Silicon oxide liner 209 advantageously prevents consumption of the gate electrode 205 by silicidation on the side surfaces thereof, and advantageously prevents a subsequently formed thin layer of NiSi on the silicon nitride sidewall spacers 211 from contacting a NiSi contact layer on the upper surface of the gate electrode 205 and/or from contacting NiSi contact layers on the upper surface of the strained Si layer 201, thereby preventing NiSi bridging along the silicon nitride sidewall spacers.

[0033] Subsequently, heavily doped source/drain regions 213b are formed in Ge-Si layer 201 using gate electrode 205 and sidewall spacers 209 and 211 as a mask, employing conventional implantation techniques, as by ion implanting boron, phosphorus, or arsenic.

[0034] Adverting to FIG. 3, a first layer of nickel 301 is deposited over the source/drain regions 213b, the spacers 211, the oxide liner 209, and the gate electrode 205. First layer of nickel 301 may be formed, for example, with a platinum content of about 8 wt. % to about 12 wt. %, such as about 10 wt. %. The first layer of nickel can be formed by sputtering using a target comprising Ni and 10 wt. % Pt. Over the first layer of nickel 301, a second layer of nickel 303 is deposited.

The second layer of nickel 303 may be formed, for example, with a platinum content of about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %. The second layer of nickel can be sputtered using a target of Ni containing an amount of Pt within the range of 30 to 100 %. The first nickel layer 301 and second nickel layer 303 may be formed to a combined thickness of about 45 Å to about 100 Å using a conventional deposition tool, for example on an AMAT Endura sputtering system, at a power of 1500W to 2500W, e.g., 2000W, and using an Argon flow of 20sccm to 40sccm, e.g., 2000sccm, or an Argon and Hydrogen mixture with a flow of 3sccm to 10sccm, e.g., 5sccm. The first layer of nickel 301 may be formed at a thickness of about 60 % to about 80 % of the combined thickness, such as about 70 % of the combined thickness.

[0035] A capping layer 305 may be formed over the second nickel layer 303 to prevent oxidation of the underlying metal. Capping layer 305 may be titanium nitride.

[0036] RTA and etch processes are performed to form a NiSi layer 401 over the source/drain regions 213b and the gate electrode 205, as illustrated in FIG. 4. The resulting NiSi layer 401 has a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the silicon interface 403 to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the opposite surface. NiSi layer 401 may be formed at a thickness of about 100 Å to about 220 Å, such as at a thickness of about 100 Å to about 150 Å.

[0037] Nickel silicide is commonly doped with platinum to improve the material and electrical properties. Platinum has proven to be effective in preventing the nickel disilicide formation (NiSi₂). It was found that strategic platinum doping can also be used to effectively shield a NiSi layer from physical damage during MOL processing (post NiSi processing) while concomitantly maintaining the NiSi integrity. However, platinum also increases the resistivity of NiSi. In accordance with embodiments of the present disclosure, effective shielding of a NiSi layer is achieved while avoiding a problematic increase in resistivity by compositionally grading a platinum doped NiSi layer.

[0038] The RTA process may be a one-step or two-step process (with a clean in between the two steps) in an inert or reducing atmosphere, e.g., a nitrogen (N₂)-containing atmosphere. The single step anneal may be implemented at a temperature of about 340° C to about 450° C, for

example at a temperature of about 360° C to about 400° C, for about 5 to 35 seconds. During the single step anneal the nickel layers are converted directly to a low resistivity monosilicide (NiSi).

[0039] The two-step anneal may be conducted by first heating at a temperature of about 280° C to about 350° C, for example about 300° C to about 320° C, e.g., 300° C, during which metal rich silicide, Ni₂Si, is formed. During the second step, the temperature may be raised to about 360° C to about 450° C, such as about 380° C to about 420° C, e.g., 420° C. During the second annealing step, the metal rich silicide is converted to monosilicide. The total time for the two step annealing can be the same as for the single step anneal process, for example 5 seconds for the first step and 30 seconds for the second step. The two-step anneal therefore reduces the time at which annealing occurs at a high temperature, thereby reducing the rough interfaces between the NiSi and the underlying Si layer.

[0040] After NiSi layer 401 is formed, the capping layer 305 and unreacted Ni on the wafer are then easily removed, as by wet chemical stripping. In an embodiment of the present disclosure, the capping layer and unreacted Ni are removed by an anisotropic sputter etching or by immersing the wafer into a solution of H₂SO₄, H₂O₂, and water (SPM) or a solution of NH₄OH, H₂O₂ and water (APM).

[0041] As illustrated in FIG. 5, the liner 209 and sidewall spacers 211 are then removed, exposing a portion of the upper surface of strained Si layer 201 between the silicide layers 401 and the side surfaces of gate electrode 205, with a very thin oxide layer (not shown) thereon, e.g., less than 50 Å, serving as a buffer layer. A highly stressed dielectric layer 601, such as a silicon nitride layer exhibiting high compressive stress, is then deposited by PECVD, as shown in FIG. 6. The highly stressed dielectric layer 601 serves to increase channel electron mobility, thereby increasing the drive current.

[0042] Another embodiment of the present disclosure is schematically illustrated in FIGS. 7 through 15. Adverting to FIG. 7, a CMOS device is schematically illustrated comprising an NMOS transistor portion at the left and a PMOS transistor portion on the right, wherein similar features are denoted by similar reference characters. Strained Si layer 701 is formed over Si-Ge layer 703. As in previous discussed embodiments, Si layer 701 may be globally strained or

locally strained in the source/drain regions. Gate electrodes 705 are formed over strained Si layer 701 with gate dielectric 707 therebetween. Lightly doped source/drain regions 713a are formed in Si layer 701 using gate electrodes 705 as a mask in a conventional manner. Sidewall spacers comprising oxide liner 709, as at a thickness of about 60 Å to about 600 Å, are formed on side surfaces of gate electrodes 705 and on portions of the upper surface of strained Si layer 701. Silicon oxide liner 709 can be formed in the same manner as discussed with respect to silicon oxide liner 209 in FIG. 2. Silicon nitride spacers 711 are then formed on silicon oxide liners 709 as by employing PECVD followed by etching.

[0043] Subsequently, heavily doped source/drain regions 713b are formed in Si layer 701 using gate electrodes 705 and sidewall spacers 709 and 711 as a mask, employing conventional implantation techniques, as by ion implanting boron, phosphorus, or arsenic.

[0044] As illustrated in FIG. 8, a first layer of nickel 801 is deposited over the source/drain regions 713b, spacers 711, oxide liners 709, and gate electrodes 705. First layer of nickel 801 may be formed, for example, with a platinum content of about 8 wt. % to about 12 wt. %, such as about 10 wt. %. Over the first layer of nickel 801, a second layer of nickel 803 is deposited. Second layer of nickel 803 may be formed, for example, with a platinum content of about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %. The first nickel layer 801 and second nickel layer 803 may be formed in a conventional deposition environment, for example on an AMAT Endura sputtering system, to a combined thickness of about 45 Å to about 100 Å. The first layer of nickel 801 may be formed at a thickness of about 60 % to about 80 % of the combined thickness, such as about 70 % of the combined thickness.

[0045] A capping layer 805 is formed over the second nickel layer 803 to prevent oxidation of the underlying metal. Capping layer 805 may be titanium nitride.

[0046] RTA and etch processes are performed to form NiSi layers 901 over source/drain regions 713b and gate electrodes 705, as illustrated in FIG. 9. The resulting NiSi layers 901 have a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the bottom surface that contacts the silicon gate electrodes 705 and source/drain regions 713b, to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the opposite surface. NiSi layers 901 may be formed at a thickness of about 100 Å to about 220 Å, such as at

a thickness of about 100 Å to about 150 Å. The RTA process may be a one-step or two-step process, as described above with respect to FIG. 4.

[0047] After NiSi layers 901 are formed, the capping layer 805 and unreacted Ni on the wafer are then easily removed, as by wet chemical stripping. In an embodiment of the present disclosure, the capping layer and unreacted Ni are removed by an anisotropic sputter etching or by immersing the wafer into a solution of H₂SO₄, H₂O₂, and water (SPM) or a solution of NH₄OH, H₂O₂ and water (APM).

[0048] As illustrated in FIG. 10, liners 709 and sidewall spacers 711 are then removed exposing portions of the upper surface of strained Si layer 701 between the silicide layers 901 and the side surfaces of gate electrodes 705, with a very thin oxide layer (not shown) thereon, e.g., less than 50 Å, serving as a buffer layer.

[0049] A highly compressive stressed silicon nitride film 1101, e.g., having a compressive stress greater than 1.5 GPa, is then deposited over both the NMOS and PMOS transistors as shown in FIG. 11. Deposition of highly compressive stressed silicon nitride film 1101 may be implemented in a conventional manner. Multiple layer deposition and plasma treatment further increases compressive stress. Subsequently, a thin oxide or oxynitride film 1103 is deposited, as illustrated in FIG. 11, by a conventional CVD process. Typically oxide or oxynitride film 1103 is deposited at a thickness of about 30 Å to about 60 Å.

[0050] Subsequently, a mask 1201, e.g., photoresist or hard mask is applied over the PMOS transistor, as illustrated in FIG. 12, and the oxide or oxynitride film 1103 and high compressive stress silicon nitride film 1101 are removed from the NMOS transistor.

[0051] Adverting to FIG. 13, the mask 1201 is removed from the PMOS transistor. A highly tensile stressed silicon nitride film 1301, having a tensile stress of greater than 1.5 GPa, is then deposited over both the PMOS and NMOS transistors. Deposition of highly tensile stressed film 1301 can be implemented in a conventional manner. A thin oxide or oxynitride film 1303 is then deposited by a conventional CVD process, as at a thickness of about 30 Å to about 60 Å.

[0052] Adverting to FIG. 14, a mask 1401, such as a photoresist or hard mask, is then applied over the NMOS transistor, and the oxide or oxynitride film 1303 and the highly tensile stressed

silicon nitride film 1301 are selectively removed from the PMOS transistor stopping on the oxide or oxynitride film 1103. The mask 1401 is then removed, and the resulting structure is illustrated in FIG. 15 which comprises an oxide or oxynitride film 1303 and a highly tensile stressed silicon nitride film 1301 over the NMOS transistor and an oxide or oxynitride film 1103 and a highly compressive stressed silicon nitride film 1101 over the PMOS transistor. The resulting CMOS device comprises both PMOS and NMOS transistors with increased channel carrier mobility and, hence, increased drive current, without increasing the resistivity of the transistors.

[0053] The embodiments of the present disclosure can achieve several technical effects, including resistance reduction, improved transistor and circuit performance, and reduced power requirements. The present disclosure enjoys industrial applicability in any of various types of highly integrated semiconductor devices.

[0054] In the preceding description, the present disclosure is described with reference to specifically exemplary embodiments thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of the present disclosure, as set forth in the claims. The specification and drawings are, accordingly, to be regarded as illustrative and not as restrictive. It is understood that the present disclosure is capable of using various other combinations and embodiments and is capable of any changes or modifications within the scope of the inventive concept as expressed herein.

WHAT IS CLAIMED IS:

1. A method comprising:
forming a transistor on a substrate, the transistor comprising source/drain regions and a gate electrode;
forming a first layer of nickel, containing a first amount of platinum, over the transistor;
forming a second layer of nickel, containing a second amount of platinum, on the first layer of nickel, wherein the second amount of platinum is greater than the first amount of platinum; and
annealing the first and second layers of nickel to form a layer of nickel silicide over the source/drain regions and the gate electrode, wherein the nickel silicide layer has a platinum composition gradient with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.
2. The method according to claim 1, comprising forming the first and second layers of nickel at a combined thickness of about from 45 Å to about 100 Å.
3. The method according to claim 2, comprising forming the first layer of nickel at a thickness of about 60 % to about 80 % of the combined thickness.
4. The method according to claim 3, comprising annealing to form the layer of nickel silicide at a thickness of about 100 Å to about 220 Å.

5. The method according to claim 1, comprising annealing by heating at a temperature of about 340° C to about 450° C.

6. The method according to claim 5, comprising annealing by heating at a temperature of about 360° C to about 400° C.

7. The method according to claim 1, comprising annealing by:

(a) heating at a first temperature of about 280° C to about 350° C;

(b) followed by heating at a second temperature of about 360° C to about 450° C.

8. The method according to claim 7, comprising heating at the first temperature of about 300° C to about 320° C and heating at the second temperature of about 380° C to about 420° C.

9. The method according to claim 1, comprising forming the first layer of nickel with a platinum content of about 8 wt. % to about 12 wt. %.

10. The method according to claim 9, comprising forming the second layer of nickel with a platinum content of about 30 wt. % to about 70 wt. %.

11. The method according to claim 10, comprising:

forming the first nickel layer of nickel at a platinum content of about 10 wt. %; and

forming the second layer of nickel at a platinum content of about 30 wt. % to about 50 wt. %.

12. The method according to claim 1, further comprising forming a capping layer of titanium nitride over the second layer of nickel before annealing.

13. The method according to claim 1, further comprising:
removing the capping layer after annealing; and
etching to remove unreacted nickel.

14. A semiconductor device comprising:
a silicon substrate,
a transistor, comprising source/drain regions and a gate electrode, on the substrate; and
a nickel silicide layer on the gate electrode and source/drain regions,
wherein the nickel silicide layer comprises platinum having a composition gradient with
platinum increasing in amount in a direction away from the source/drain regions and the
gate electrode.

15. The semiconductor device according to claim 14, further comprising:
a P-channel transistor;
an N-channel transistor;
a first nickel silicide layer on the P-channel transistor;
a second nickel silicide layer on the N-channel transistor;
wherein each of the first and second nickel silicide layers comprises platinum having a
concentration gradient with platinum increasing in amount in a direction away from the
P-channel and N-channel transistors;

a first stress liner on the P-channel transistor; and
a second stress liner on the N-channel transistor.

16. The semiconductor device according to claim 14, wherein the concentration gradient of platinum in the nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

17. The semiconductor device according to claim 16, wherein the concentration gradient of platinum in the nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

18. The semiconductor device according to claim 14, wherein the nickel silicide layer has a thickness of about 100 Å to about 220 Å.

19. The semiconductor device according to claim 14, wherein the nickel silicide layer has a thickness of about 100 Å to about 150 Å.

20. A method comprising:

forming a P-channel and an N-channel transistor on a silicon substrate, each transistor including a polysilicon gate electrode and source/drain regions formed in the silicon substrate;
depositing a first layer of nickel containing a first amount of platinum on both transistors;

depositing a second layer of nickel having a second amount of platinum on the first layer of nickel, wherein the second amount of platinum is greater than the first amount of platinum;

depositing a titanium nitride capping layer on the second layer of nickel;

annealing the first and second layers of nickel to form a layer of nickel silicide containing platinum having a concentration gradient with platinum increasing in amount in a direction away from the gate electrode and source/drain regions,

etching to remove the capping layer and unreacted nickel;

forming a first stress liner exhibiting high compressive stress over both transistors;

selectively removing the first stress liner from the N-channel transistor;

forming a second stress liner exhibiting high tensile stress over both transistors; and

selectively removing the second stress liner from the P-channel transistor.

ABSTRACT OF DISCLOSURE

Transistor devices are formed with nickel silicide layers formulated to prevent degradation upon removal of overlying stress liners. Embodiments include transistors with nickel silicide layers having a platinum composition gradient increasing in platinum content toward the upper surfaces thereof, i.e., increasing in platinum in a direction away from the gate electrode and source/drain regions. Embodiments include forming a first layer of nickel having a first amount of platinum and forming, on the first layer of nickel, a second layer of nickel having a second amount of platinum, the second weight percent of platinum being greater than the first weight percent. The layers of nickel are then annealed to form a nickel silicide layer having the platinum composition gradient increasing in platinum toward the upper surface. The platinum concentration gradient protects the nickel silicide layer during subsequent processing, as during etching to remove overlying stress liners, thereby avoiding a decrease in device performance.

Electronic Patent Application Fee Transmittal				
Application Number:				
Filing Date:				
Title of Invention:		METHOD TO REDUCE MOL DAMAGE ON NiSi		
First Named Inventor/Applicant Name:		Karthik Ramani		
Filer:		Arthur J. Steiner/Brandy Stoeckel		
Attorney Docket Number:		P2343US01		
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	330	330
Utility Search Fee	1111	1	540	540
Utility Examination Fee	1311	1	220	220
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1090

Electronic Acknowledgement Receipt

EFS ID:	9973314
Application Number:	13096511
International Application Number:	
Confirmation Number:	4709
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NiSi
First Named Inventor/Applicant Name:	Karthik Ramani
Customer Number:	99963
Filer:	Arthur J. Steiner/Brandy Stoeckel
Filer Authorized By:	Arthur J. Steiner
Attorney Docket Number:	P2343US01
Receipt Date:	28-APR-2011
Filing Date:	
Time Stamp:	16:13:54
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1090
RAM confirmation Number	2931
Deposit Account	504213
Authorized User	DITTHAVONG,PHOUPHANOMKETH

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

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GF-ITC-00004352

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Change of Address	NY0281-D1_P2343US01_ChangeofCorrespondence.pdf	195304 2701338c2de8bbdbb8989e7d969a959281aad89c	no	2
Warnings:					
Information:					
2	Application Data Sheet	NY0281-D1_P2343US01_ADS.pdf	74514 0ee7c7f19676d1dc3926b7616f1ea9cfe0a6676c	no	3
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
3	Drawings-only black and white line drawings	NY0281-D1_P2343US01_Drawings.pdf	346083 f2b5943ae12a4178c54235a82e7f7cc2feb877c3	no	13
Warnings:					
Information:					
4		NY0281-D1_P2343US01_PreliminaryAmendment.pdf	114685 f4cd171cfbea121558260f6625026af09a3a7cd8	yes	8
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Preliminary Amendment		1	1	
	Claims		2	6	
	Applicant Arguments/Remarks Made in an Amendment		7	8	
Warnings:					
Information:					
5	Oath or Declaration filed	NY0281-D1_P2343US01_Declaration.pdf	414302 e2a5402509fce23157be284b75922e784e08e49f	no	6
Warnings:					
The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing					
Information:					
6	Transmittal Letter	NY0281-D1_P2343US01_IDSTransmittalLetter.pdf	108277 31986b2aa94abc0e76c28f5c647cf72bb92fae43	no	2
Warnings:					
Information:					

7	Information Disclosure Statement (IDS) Filed (SB/08)	NY0281- D1_P2343US01_IDS_SB-08.pdf	1492933 356aeb474039470fb65087c33df7a9b732e a5e53	no	1
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
8		NY0281- D1_P2343US01_Specification. pdf	154457 87017a26ab97e976e8db61c89ee31c7c1fd 34502	yes	20
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	14	
	Claims		15	19	
	Abstract		20	20	
Warnings:					
Information:					
9	Fee Worksheet (PTO-875)	fee-info.pdf	32851 8372fa41095dbec88924c77b0a8c0a12e5f9 f5cb	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			2933406		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

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**CHANGE OF
CORRESPONDENCE ADDRESS**
*Application*Address to:
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Application Number	Not yet assigned
Filing Date	April 28, 2011
First Named Inventor	Karthik RAMANI
Art Unit	Not yet assigned
Examiner Name	Not yet assigned
Attorney Docket Number	P2343US01

Please change the Correspondence Address for the above-identified patent application to:

The address associated with
Customer Number:

99963

ORFirm or
Individual Name

Address

City

State

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I am the:



Applicant/Inventor

Assignee of record of the entire interest.
Statement under 37 CFR 3.73(b) is enclosed. (Form PTO/SB/96).Attorney or agent of record. Registration Number 26106.

Registered practitioner named in the application transmittal letter in an application without an executed oath or declaration. See 37 CFR 1.33(a)(1). Registration Number _____.

Signature

/Arthur J. Steiner/

Typed or Printed
Name

Arthur J. Steiner

Date April 28, 2011

Telephone (703) 519-9954

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below.



*Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The information provided by you in this form will be subject to the following routine uses:

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4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

APPLICATION DATA SHEET

Application Information

Application Type:: Regular
Subject Matter:: Utility
Title:: METHOD TO REDUCE MOL DAMAGE ON
NiSi
Attorney Docket Number:: P2343US01
Total Drawing Sheets:: 13
Small Entity?:: No

Inventor Information

Inventor One Authority Type:: Inventor
Primary Citizenship Country:: India
Status:: Full Capacity
Given Name:: Karthik
Family Name:: RAMANI
City of Residence:: Hopewell Junction
State or Province of Residence:: NY
Country of Residence:: USA
Street:: 3603 Chelsea Cove South
City:: Hopewell Junction
State or Province:: NY
Country:: USA
Postal or Zip Code:: 12533

Inventor Two Authority Type:: Inventor
Primary Citizenship Country:: United States
Status:: Full Capacity
Given Name:: Paul
Middle Name:: R.

Family Name:: BESSER
City of Residence:: Sunnyvale
State or Province of Residence:: CA
Country of Residence:: USA
Street:: 1087 Yorktown Drive
City:: Sunnyvale
State or Province:: CA
Country:: USA
Postal or Zip Code:: 94087

Correspondence Information

Correspondence Customer Number:: 99963

Representative Information

Representative Customer Number:: 99963

Domestic Priority Information

Application::	Continuity Type::	Parent Application::	Parent Filing Date::
This Application	Divisional of	12/366,378	February 5, 2009

Foreign Priority Information

Country::	Application Number::	Filing Date::	Priority Claimed::
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Assignee Information

Assignee Name:: GLOBALFOUNDRIES Inc.
Street:: Maples Corporate Services Limited
P.O. Box 309, Ugland House
City:: Grand Cayman
Country:: Cayman Islands
Postal or Zip Code:: KY1-1104

Signature:

Signature /Arthur J. Steiner/ Date (YYYY-MM-DD) 2011-04-28

First Name: Arthur J. Last Name: Steiner

Registration Number: 26106

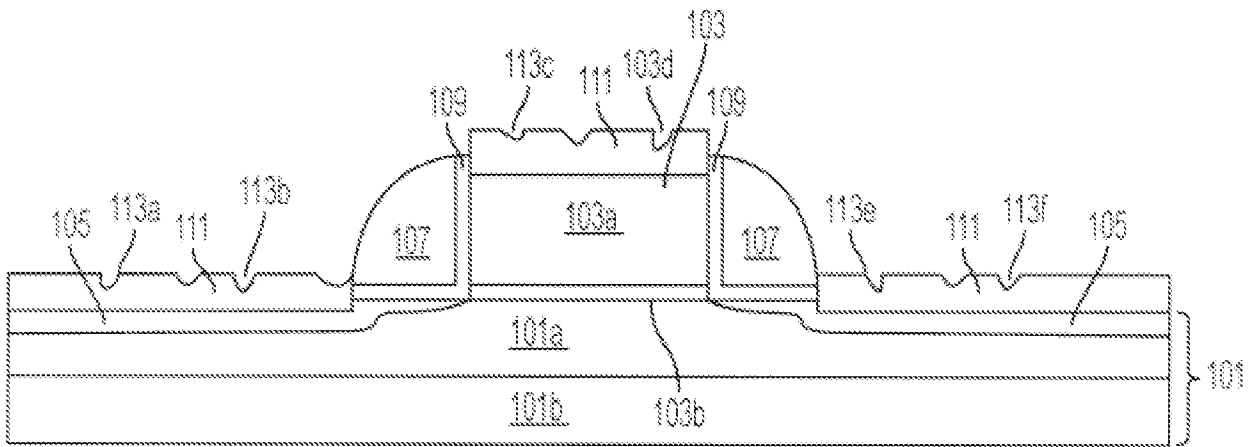


FIG. 1
(PRIOR ART)

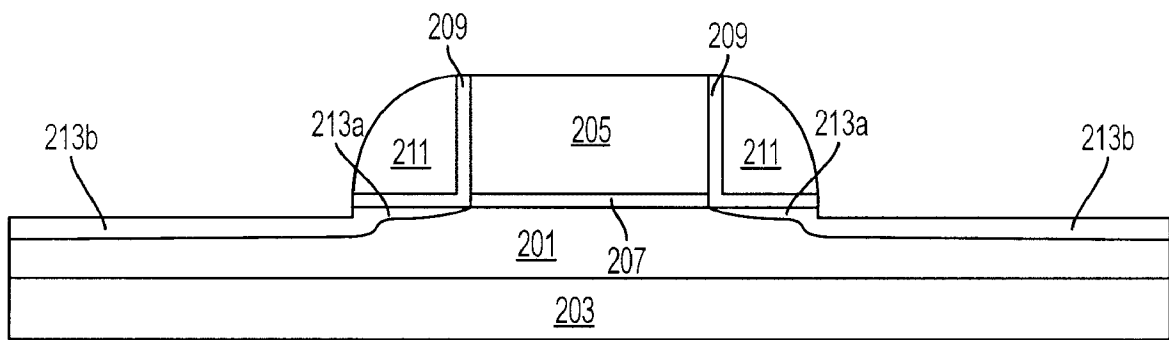


FIG. 2

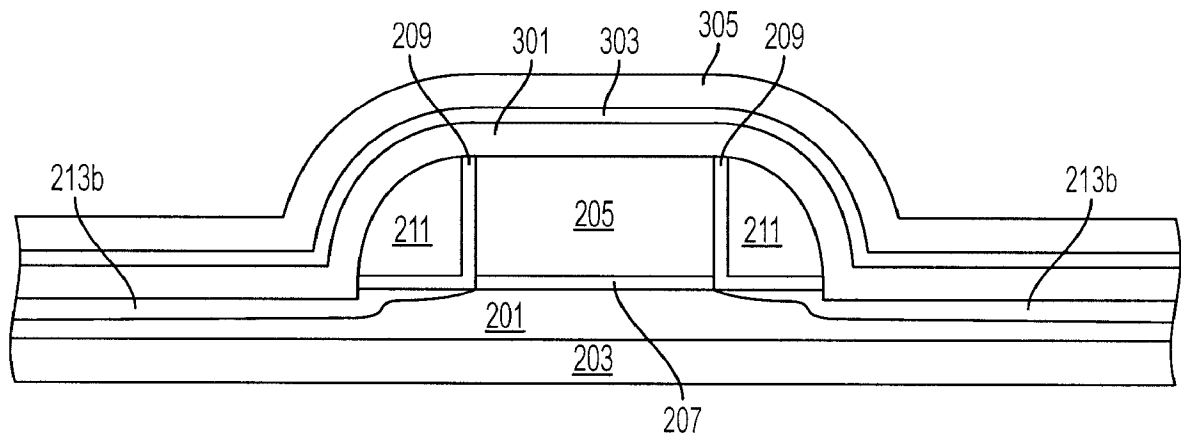


FIG. 3

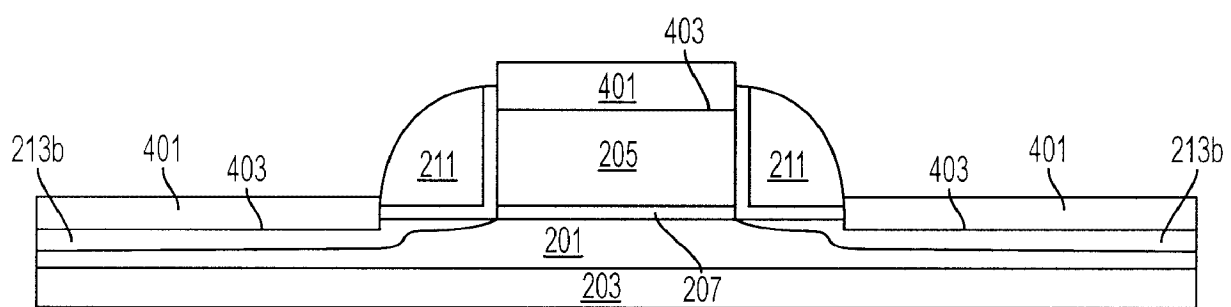


FIG. 4

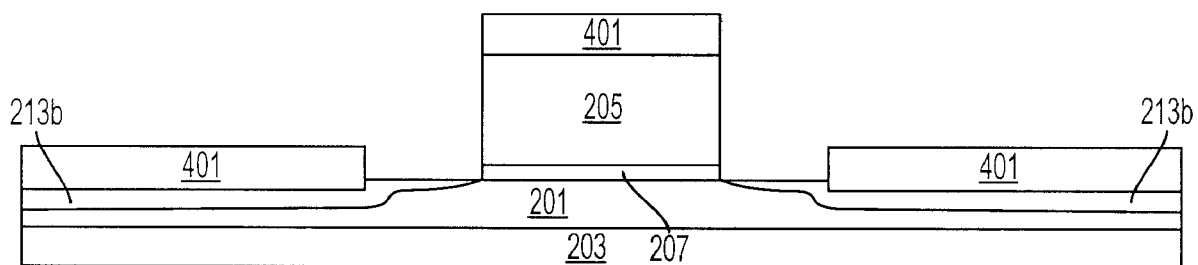


FIG. 5

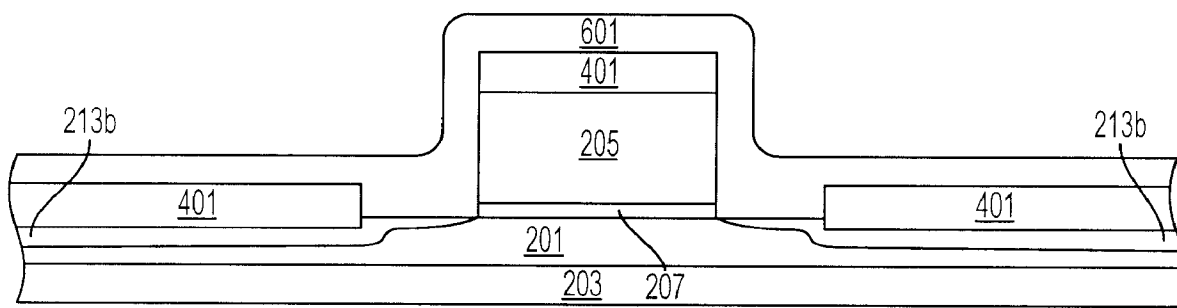


FIG. 6

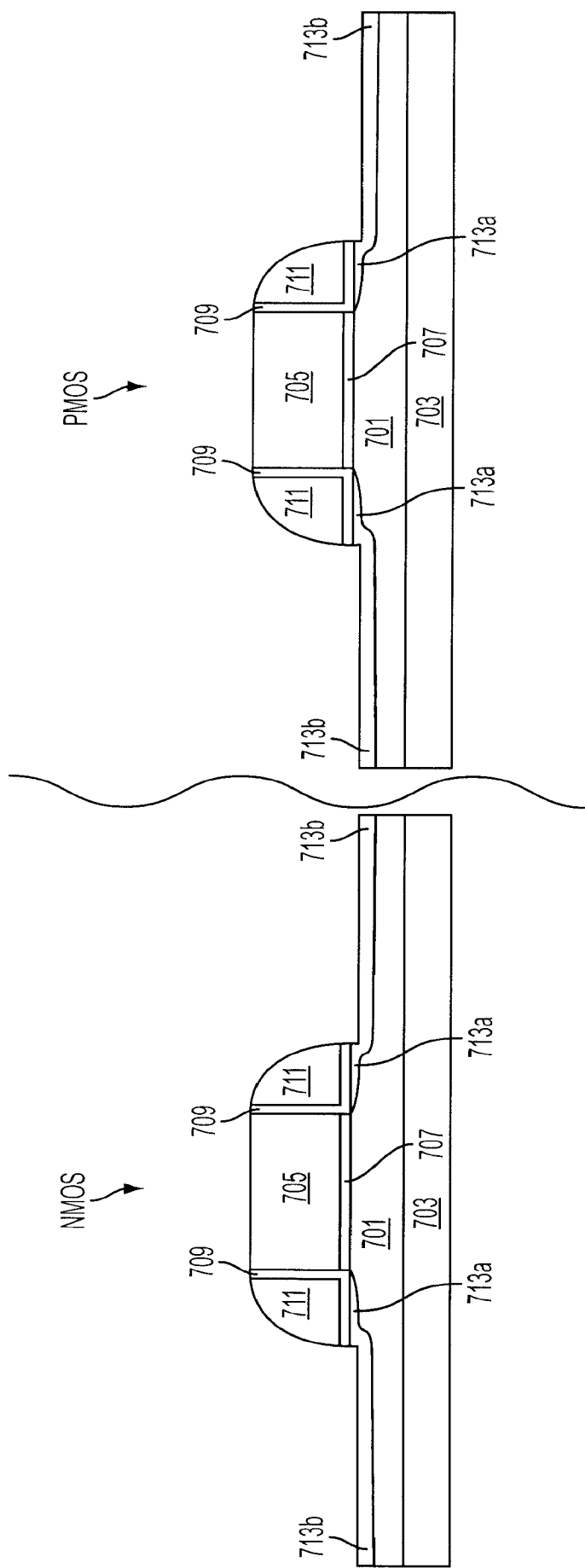


FIG. 7

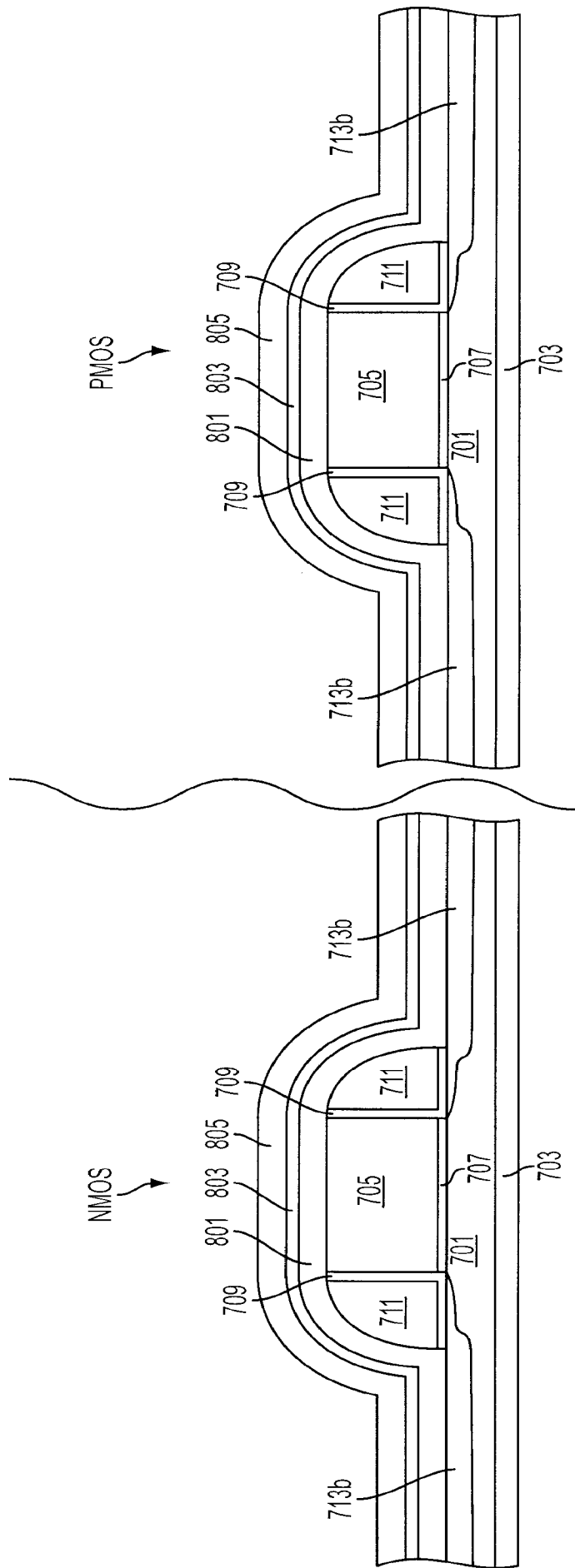


FIG. 8

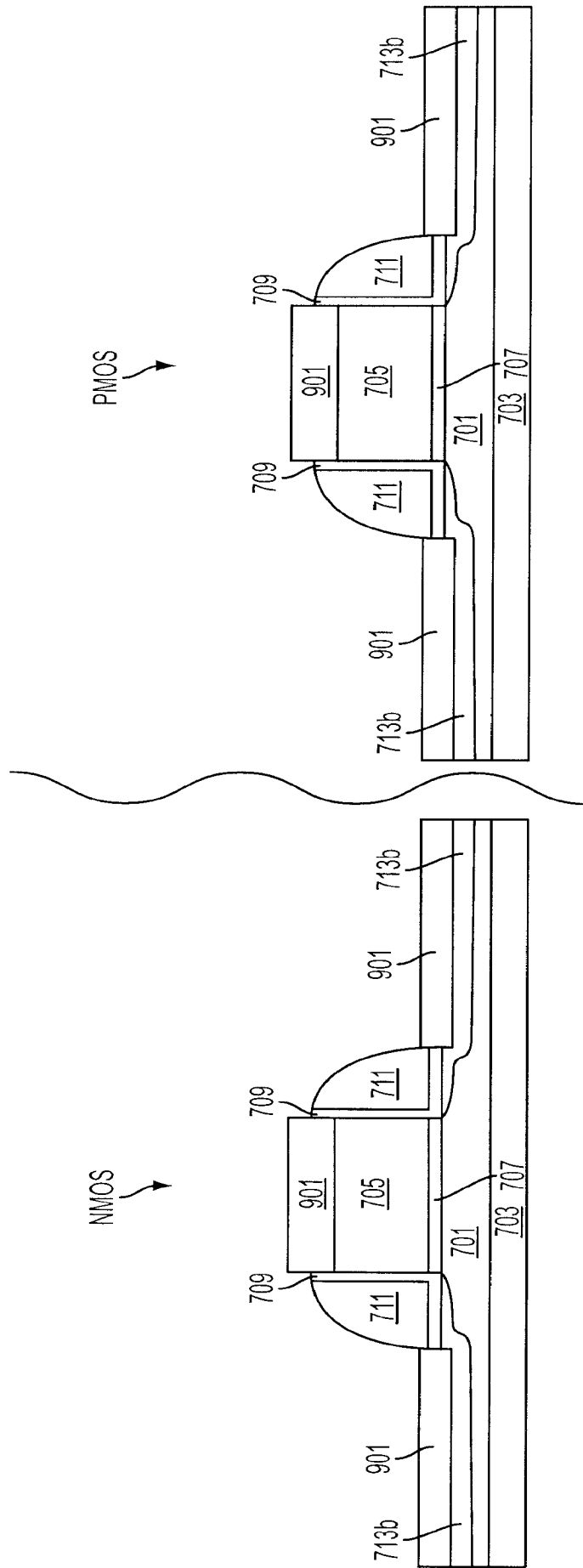


FIG. 9

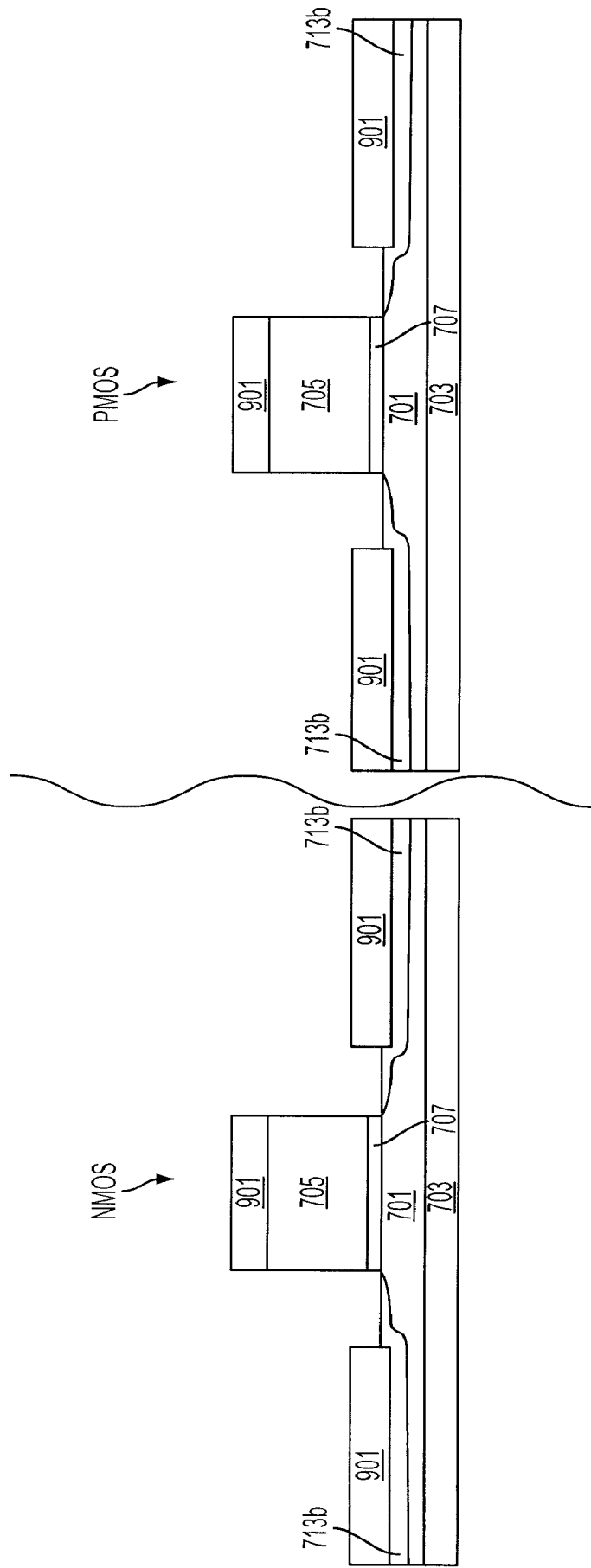


FIG. 10

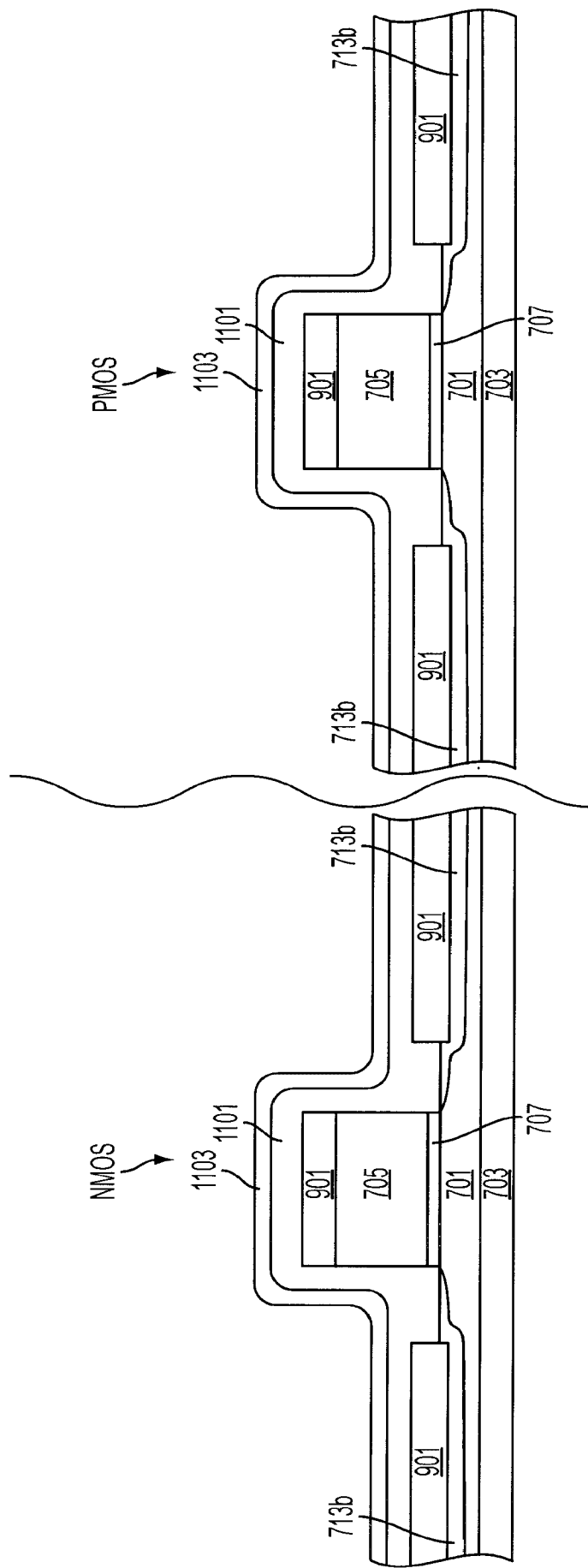


FIG. 11

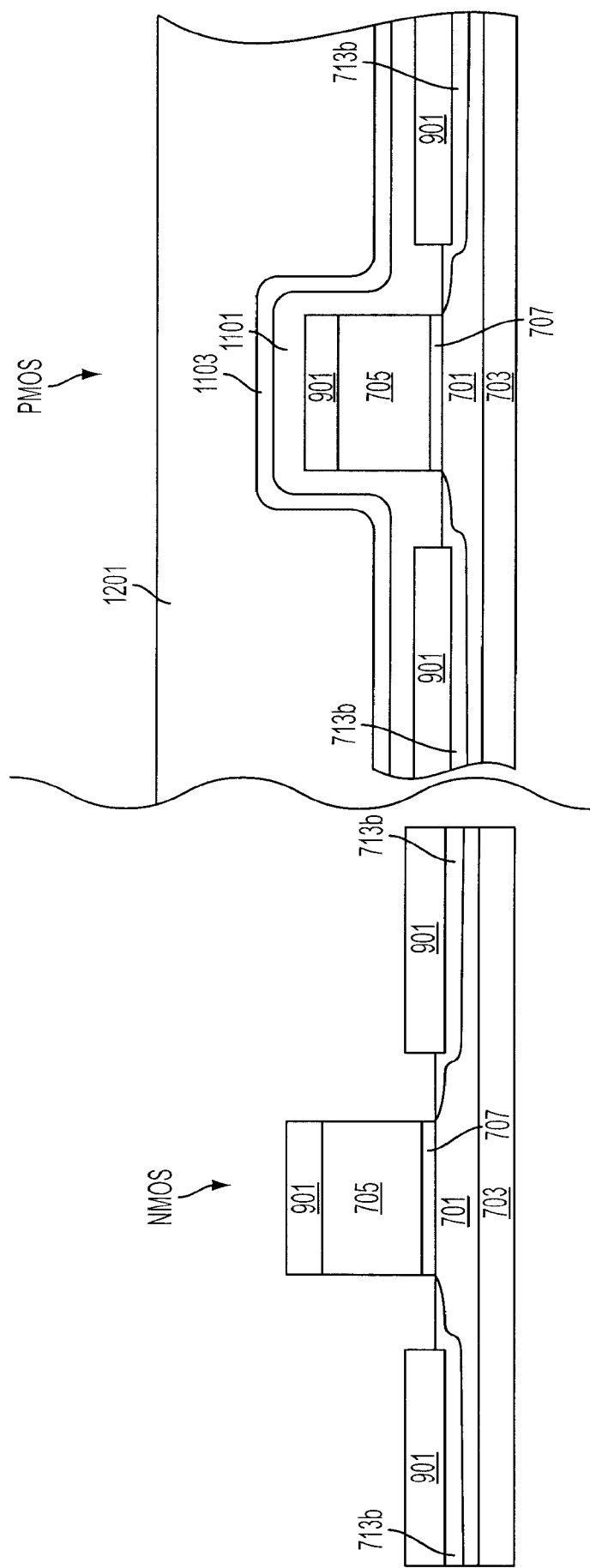


FIG. 12

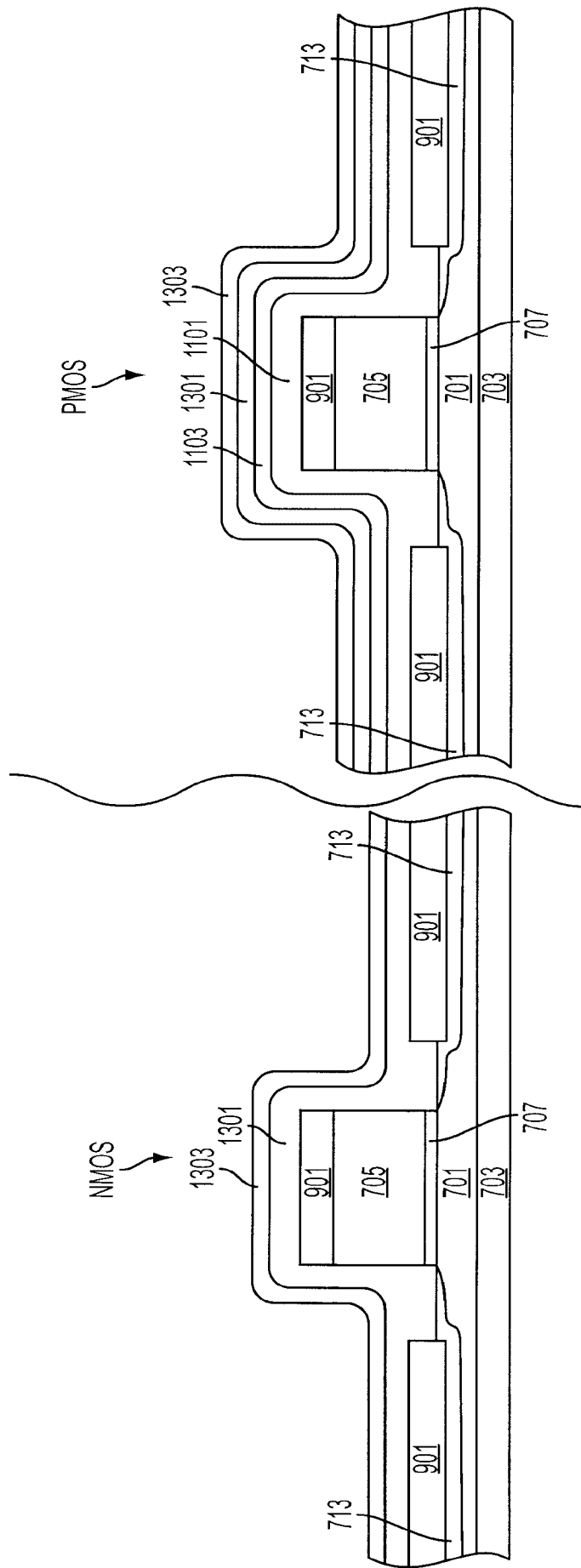


FIG. 13

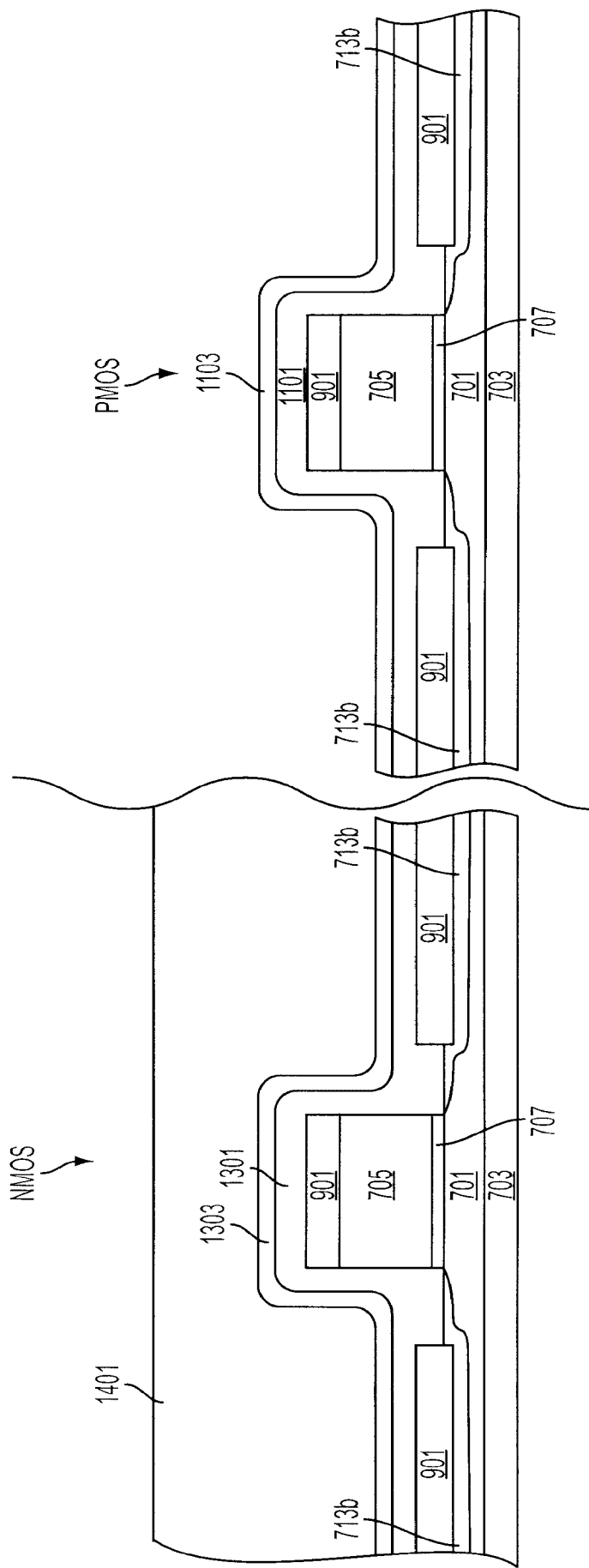


FIG. 14

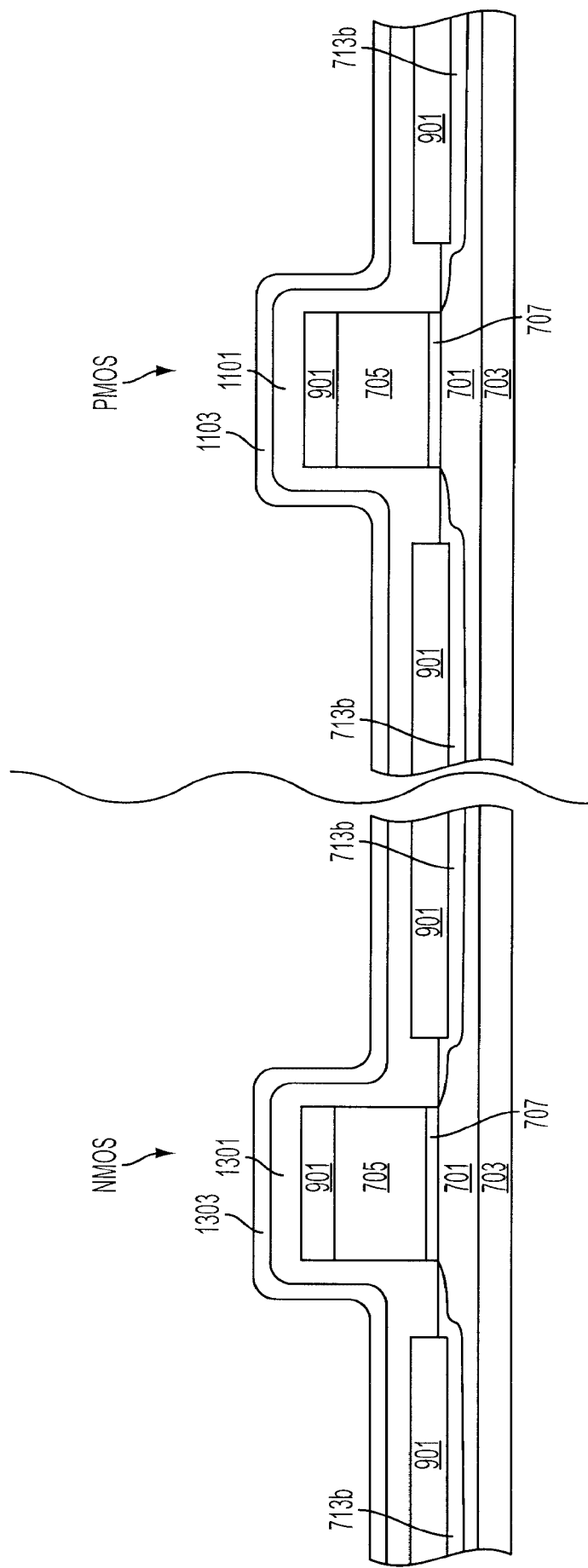


FIG. 15



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/096,511	04/28/2011	2812	1090	P2343US01	20	3

CONFIRMATION NO. 4709

99963

DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

FILING RECEIPT



OC000000047597001

Date Mailed: 05/12/2011

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Karthik Ramani, Hopewell Junction, NY;
Paul R. BESSER, Sunnyvale, CA;

Assignment For Published Patent Application

GLOBALFOUNDRIES Inc., Grand Cayman, CAYMAN ISLANDS

Power of Attorney: The patent practitioners associated with Customer Number 30671

Domestic Priority data as claimed by applicant

This application is a DIV of 12/366,378 02/05/2009

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

If Required, Foreign Filing License Granted: 05/10/2011

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/096,511**

Projected Publication Date: 08/18/2011

Non-Publication Request: No

Early Publication Request: No

Title

METHOD TO REDUCE MOL DAMAGE ON NiSi

Preliminary Class

438

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For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875						Application or Docket Number 13/096,511				
APPLICATION AS FILED - PART I										
(Column 1)		(Column 2)		SMALL ENTITY		OR OTHER THAN SMALL ENTITY				
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)			
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	330			
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	540			
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	220			
TOTAL CLAIMS (37 CFR 1.16(j))	20	minus 20 = *			OR	x 52 =	0.00			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3	minus 3 = *				x 220 =	0.00			
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$270 (\$135 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00			
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00			
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1090			
APPLICATION AS AMENDED - PART II										
(Column 1)		(Column 2)		(Column 3)		SMALL ENTITY		OR OTHER THAN SMALL ENTITY		
AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
(Column 1)		(Column 2)		(Column 3)		SMALL ENTITY		OR OTHER THAN SMALL ENTITY		
AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.										

To: docket@dcpatent.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 99963

May 12, 2011 05:32:49 AM

Dear PAIR Customer:

DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314
UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 99963 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

Disclaimer:

The list of documents shown below is provided as a courtesy and is not part of the official file wrapper. The content of the images shown in PAIR is the official record.

Application	Document	Mailroom Date	Attorney Docket No.
13096511	APP.FILE.REC	05/12/2011	P2343US01

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>.

If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov with 'e-Office Action' on the subject line or call 1-866-217-9197 during the following hours:

Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:	Karthik RAMANI <i>et al.</i>	Confirmation No.:	4709
Application No.:	13/096,511	Examiner:	Not yet assigned
Filed:	April 28, 2011	Group Art Unit:	2813

For: METHOD TO REDUCE MOL DAMAGE ON NiSi

Commissioner for Patents
Alexandria, VA 22313-1450

REQUEST FOR CORRECTED FILING RECEIPT

Dear Sir:
Office of Initial Patent Examinations
Customer Service Center
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Petitioner hereby requests a correction in the Filing Receipt of the above-referenced patent application.

Power of Attorney:

The patent practitioners associated with Customer Number ~~30671~~

Power of Attorney:

The patent practitioners associated with Customer Number 99963

Enclosed is a copy of the Filing Receipt with the changes noted thereon.

Respectfully submitted,

DITTHAVONG MORI & STEINER, P.C.

May 12, 2011
Date

/Arthur J. Steiner/
Arthur J. Steiner
Attorney/Agent for Applicant(s)
Reg. No. 26106



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/096,511	04/28/2011	2812	1090	P2343US01	20	3

CONFIRMATION NO. 4709

99963

DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

FILING RECEIPT



OC000000047597001*

Date Mailed: 05/12/2011

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Karthik Ramani, Hopewell Junction, NY;
Paul R. BESSER, Sunnyvale, CA;

Assignment For Published Patent Application

GLOBALFOUNDRIES Inc., Grand Cayman, CAYMAN ISLANDS

Power of Attorney: The patent practitioners associated with Customer Number ~~30671~~

Domestic Priority data as claimed by applicant

This application is a DIV of 12/366,378 02/05/2009

Power of Attorney: The patent practitioners associated
with Customer Number 99963

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

If Required, Foreign Filing License Granted: 05/10/2011

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/096,511**

Projected Publication Date: 08/18/2011

Non-Publication Request: No

Early Publication Request: No

Title

METHOD TO REDUCE MOL DAMAGE ON NiSi

Preliminary Class

438

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER**Title 35, United States Code, Section 184****Title 37, Code of Federal Regulations, 5.11 & 5.15****GRANTED**

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as

set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

Electronic Acknowledgement Receipt

EFS ID:	10074292
Application Number:	13096511
International Application Number:	
Confirmation Number:	4709
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NiSi
First Named Inventor/Applicant Name:	Karthik Ramani
Customer Number:	99963
Filer:	Arthur J. Steiner/Brandy Stoeckel
Filer Authorized By:	Arthur J. Steiner
Attorney Docket Number:	P2343US01
Receipt Date:	12-MAY-2011
Filing Date:	28-APR-2011
Time Stamp:	16:43:55
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		NY0281-D1_P2343US01_RequestforCorrectedFilingReceipt_filed05-12-11.pdf	257190 e8ae037852bcc5ae2ac1b052af83fd552821209	yes	4

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Request for Corrected Filing Receipt	1	1
	Request for Corrected Filing Receipt	2	4
Warnings:			
Information:			
Total Files Size (in bytes):		257190	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.			



UNITED STATES PATENT AND TRADEMARK OFFICE

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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/096,511	04/28/2011	Karthik Ramani	P2343US01

CONFIRMATION NO. 4709

IMPROPER CPOA LETTER

99963
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314



Date Mailed: 05/19/2011

NOTICE REGARDING POWER OF ATTORNEY

This is in response to the Power of Attorney filed 05/12/2011. The Power of Attorney in this application is not accepted for the reason(s) listed below:

- The Power of Attorney you provided did not designate the Power of Attorney to your Customer Number. If you would like the Power of Attorney to be associated with the practitioners associated with the Customer Number, a new Power of Attorney must be provided revoking the Power of Attorney that has been established and designating the use of the Customer Number for Power of Attorney.

/ttran/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

To: docket@dcpatent.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 99963

May 19, 2011 05:30:52 AM

Dear PAIR Customer:

DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314
UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 99963 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

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Disclaimer:

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Application	Document	Mailroom Date	Attorney Docket No.
13096511	N572	05/19/2011	P2343US01

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>.

If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov with 'e-Office Action' on the subject line or call 1-866-217-9197 during the following hours:

Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Application of:	Karthik RAMANI <i>et al.</i>	Confirmation No.:	4709
Application No.:	13/096,511	Examiner:	Not yet assigned
Filed:	April 28, 2011	Group Art Unit:	2813

For: METHOD TO REDUCE MOL DAMAGE ON NiSi

Commissioner for Patents
Alexandria, VA 22313-1450

CHANGE OF CORRESPONDENCE/POWER OF ATTORNEY

Dear Sir:

Enclosed please find the PTO/SB/80 Power of attorney form for prosecuting the above-mentioned application before the USPTO to be associated with Customer No. 99963.

Also enclosed is Assignee showing of ownership per Statement under 37 CFR 3.73(b) for the application no. 12/366,378. The Divisional Application filed April 28, 2011 is a divisional of U.S. Application Serial No. 12/366,378, filed on February 5, 2009.

Favorable consideration and acceptance of Power of attorney is respectfully requested. If any unresolved issues remain, it is respectfully requested that the Examiner telephone the undersigned attorney at (703) 519-9954 so that such issues may be resolved as expeditiously as possible.

To the extent necessary, a petition for an extension of time under 37 C.F.R. § 1.136 is hereby made. Please charge any shortage in fees due in connection with the filing of this paper, including extension of time fees, to Deposit Account 504213 and please credit any excess fees to such deposit account.

Respectfully submitted,
DITTHAVONG MORI & STEINER, P.C.

May 19, 2011
Date

/Arthur J. Steiner/
Arthur J. Steiner
Attorney/Agent for Applicant(s)
Reg. No. 26106

POWER OF ATTORNEY TO PROSECUTE APPLICATIONS BEFORE THE USPTO

I hereby revoke all previous powers of attorney given in the application identified in the attached statement under 37 CFR 3.73(b).

I hereby appoint:

☒ Practitioners associated with the Customer Number:

99963

OR

☐ Practitioner(s) named below (if more than ten patent practitioners are to be named, then a customer number must be used):

Name	Registration Number	Name	Registration Number

as attorney(s) or agent(s) to represent the undersigned before the United States Patent and Trademark Office (USPTO) in connection with any and all patent applications assigned only to the undersigned according to the USPTO assignment records or assignment documents attached to this form in accordance with 37 CFR 3.73(b).

Please change the correspondence address for the application identified in the attached statement under 37 CFR 3.73(b) to:

☒ The address associated with Customer Number:

99963

OR

☐ Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone			Email

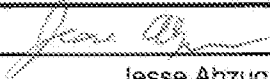
Assignee Name and Address:

GLOBALFOUNDRIES INC.
Maples Corporate Services Limited
P.O. Box 309, Ugland House
Grand Cayman, Cayman Islands KY1-1104

A copy of this form, together with a statement under 37 CFR 3.73(b) (Form PTO/SB/96 or equivalent) is required to be filed in each application in which this form is used. The statement under 37 CFR 3.73(b) may be completed by one of the practitioners appointed in this form if the appointed practitioner is authorized to act on behalf of the assignee, and must identify the application in which this Power of Attorney is to be filed.

SIGNATURE of Assignee of Record

The individual whose signature and title is supplied below is authorized to act on behalf of the assignee

Signature		Date	2 DEC 2010
Name	Jesse Abzug, Esq.	Telephone	518-305-9166
Title	Senior Counsel, IP Law		

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

GF-ITC-00004387

STATEMENT UNDER 37 CFR 3.73(b)Applicant/Patent Owner: Karthik Ramani et al.Application No./Patent No.: 12/366,378Filed/Issue Date: February 5, 2009Titled: METHOD TO REDUCE MOL DAMAGE ON NISI

GLOBALFOUNDRIES INC., a Corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that it is:

1. ☒ the assignee of the entire right, title, and interest in;
2. ☐ an assignee of less than the entire right, title, and interest in
(The extent (by percentage) of its ownership interest is _____ %); or
3. ☐ the assignee of an undivided interest in the entirety of (a complete assignment from one of the joint inventors was made)

the patent application/patent identified above, by virtue of either:

- A. ☐ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel _____, Frame _____, or for which a copy therefore is attached.

OR

- B. ☒ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: Karthik Ramani et al. To: ADVANCED MICRO DEVICES, INC.

The document was recorded in the United States Patent and Trademark Office at
Reel 022213, Frame 0395, or for which a copy thereof is attached.

2. From: ADVANCED MICRO DEVICES, INC. To: GLOBALFOUNDRIES INC.

The document was recorded in the United States Patent and Trademark Office at
Reel 023120, Frame 0426, or for which a copy thereof is attached.

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

- ☒ As required by 37 CFR 3.73(b)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

Signature

Date

Jesse Abzug, Esq.

Senior Counsel, IP Law

Printed or Typed Name

Title

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

GF-ITC-00004388

Electronic Acknowledgement Receipt

EFS ID:	10127549
Application Number:	13096511
International Application Number:	
Confirmation Number:	4709
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NiSi
First Named Inventor/Applicant Name:	Karthik Ramani
Customer Number:	99963
Filer:	Arthur J. Steiner/Brandy Stoeckel
Filer Authorized By:	Arthur J. Steiner
Attorney Docket Number:	P2343US01
Receipt Date:	19-MAY-2011
Filing Date:	28-APR-2011
Time Stamp:	15:35:47
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		NY0281-D1_P2343US01_SB80_SB96.pdf	730660 2b588c3a6272bb96d30da1743e557542ea5647e9	yes	3

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Power of Attorney	1	2
	Assignee showing of ownership per 37 CFR 3.73(b).	3	3
Warnings:			
Information:			
Total Files Size (in bytes):		730660	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.			



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/096,511	04/28/2011	Karthik Ramani	P2343US01

99963
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

CONFIRMATION NO. 4709
POA ACCEPTANCE LETTER



Date Mailed: 06/01/2011

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 05/19/2011.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/hsarwari/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/096,511	04/28/2011	Karthik Ramani	P2343US01

99963
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

CONFIRMATION NO. 4709
POWER OF ATTORNEY NOTICE



Date Mailed: 06/01/2011

NOTICE REGARDING CHANGE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 05/19/2011.

- The Power of Attorney to you in this application has been revoked by the assignee who has intervened as provided by 37 CFR 3.71. Future correspondence will be mailed to the new address of record(37 CFR 1.33).

/hsarwari/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

To: docket@dcpatent.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 99963

Jun 01, 2011 05:29:13 AM

Dear PAIR Customer:

DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314
UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No.
13096511	N570	06/01/2011	P2343US01
	N570	06/01/2011	P2343US01

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/096,511	04/28/2011	Karthik Ramani	P2343US01

CONFIRMATION NO. 4709

99963
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

PUBLICATION NOTICE



OC000000049391905

Title:METHOD TO REDUCE MOL DAMAGE ON NiSi

Publication No.US-2011-0198670-A1

Publication Date:08/18/2011

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

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Aug 19, 2011 05:20:49 AM

Dear PAIR Customer:

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918 PRINCE STREET
ALEXANDRIA, VA 22314
UNITED STATES

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13096511	NTC.PUB	08/18/2011	P2343US01

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/096,511	04/28/2011	Karthik Ramani	P2343US01	4709
99963 7590 04/26/2012 DITTHAVONG MORI & STEINER, P.C. 918 PRINCE STREET ALEXANDRIA, VA 22314			EXAMINER NICELY, JOSEPH C	
			ART UNIT 2813	PAPER NUMBER
			NOTIFICATION DATE 04/26/2012	DELIVERY MODE ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 13/096,511	Applicant(s) RAMANI ET AL.	
	Examiner Joseph C. Nicely	Art Unit 2813	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 28 April 2011.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 5) ☒ Claim(s) 14, 16-19 and 21-35 is/are pending in the application.
- 5a) Of the above claim(s) ____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☒ Claim(s) 14, 16-19 and 21-35 is/are rejected.
- 8) ☐ Claim(s) ____ is/are objected to.
- 9) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

Application Papers

- 10) ☒ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 28 April 2011 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 12) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 13) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☒ Notice of References Cited (PTO-892) | 4) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. ____. |
| 2) ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) | 5) ☐ Notice of Informal Patent Application |
| 3) ☒ Information Disclosure Statement(s) (PTO/SB/08)
Paper No(s)/Mail Date <u>04/28/2011</u> . | 6) ☐ Other: ____. |

DETAILED ACTION

1. Examiner acknowledges receipt of the Preliminary Amendment filed 04/28/2011 in which claims 1-13, 15, and 20 were cancelled and claims 21-35 were added.
2. Claims 14, 16-19, and 21-35 are currently pending and presented for examination.

Specification

3. The disclosure is objected to because of the following informalities:
 - 1) In [0006] of the application as filed, “compressive strain..” should be amended to read –compressive strain.–.
 - 2) In view of claims 21, 32, and 35, which states that a Si-Ge layer is formed on a silicon layer (silicon substrate):

In paragraphs [0042] and [0043] in the application as filed, “Strained Si layer 701” and “Si layer 701” should be amended to read –Strained Si-Ge layer 701– and –Si-Ge layer 701– and “Si-Ge layer 703” should be amended to read –Si layer 703–.
 - 3) In [0049] in the application as filed, “e..g.” should be amended to read –e.g.–.

Appropriate correction is required.

Claim Objections

4. Claim 25 is objected to because of the following informalities: in line 2, “substrate,” should be amended to read –substrate;–. Appropriate correction is required.
5. Claim 35 is objected to because of the following informalities:

In line 4, “electrode,” should be amended to read –electrode–.

In line 6, "electrode," should be amended to read –electrode–.

Appropriate correction is required.

Claim Rejections - 35 USC § 102

6. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(b) the invention was patented or described in a printed publication in this or a foreign country or in public use or on sale in this country, more than one year prior to the date of application for patent in the United States.

7. Claim 14 is rejected under 35 U.S.C. 102(b) as being anticipated by Okuno et al (US 2007/0093047 and Okuno hereinafter).

Okuno discloses **a semiconductor device comprising: a silicon substrate** (Fig. 21; [0143]; silicon substrate 100), **a transistor, comprising source/drain regions (109A) and a gate electrode (103A), on the substrate** (Fig. 21; [0144]); **and a nickel silicide layer (502) on the gate electrode and source/drain regions** (Fig. 21; [0144]), **wherein the nickel silicide layer comprises platinum having a composition gradient with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode** (Fig. 21; [0130] and [0147]; platinum containing region 505 of nickel silicide layer 502; the composition increases from the lower part of 505 toward to the top of 505).

Claim Rejections - 35 USC § 103

8. The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

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(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

9. This application currently names joint inventors. In considering patentability of the claims under 35 U.S.C. 103(a), the examiner presumes that the subject matter of the various claims was commonly owned at the time any inventions covered therein were made absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and invention dates of each claim that was not commonly owned at the time a later invention was made in order for the examiner to consider the applicability of 35 U.S.C. 103(c) and potential 35 U.S.C. 102(e), (f) or (g) prior art under 35 U.S.C. 103(a).

10. Claims 16 and 17 are rejected under 35 U.S.C. 103(a) as being unpatentable over Okuno in view of Dussarrat et al (US 2010/0256405 and Dussarrat hereinafter) in view of Lee et al ("Fully Silicided $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ Metal Gate Electrode for p-MOSFETs) in view of Cabral et al (US 2008/0274611 and Cabral hereinafter).

As to claims 16 and 17: Okuno in paragraph 7 discloses that the region 505 (platinum containing region) of the nickel silicide layer 502 has a gradient that changes (increases) from the bottom of 505 toward the top of 505.

Okuno fails to expressly disclose the concentration of platinum within the layer 505.

Therefore, Okuno fails to expressly disclose **wherein the concentration gradient of platinum in the nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %; wherein the**

concentration gradient of platinum in the nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

Dussarrat discloses that platinum doping in the range of 5-10 wt. % are used in nickel silicide formation (see [0006]). Lee discloses in the Abstract that a 5-33 wt. % for platinum in a nickel silicide for p-MOSFETs is well known. Cabral discloses that n-MOSFETs can also use nickel silicides containing platinum is well known (see [0073]). Thus, Lee in view of Cabral teaches that nickel silicides containing platinum can be applied to CMOS applications as both NMOS and PMOS devices can use the silicide layers. Therefore, platinum concentrations within both claimed ranges are used and well known in the prior art.

As Okuno discloses that the platinum concentration increases from a lower portion of the layer 505 toward the top surface, and platinum concentrations within both (the upper and lower ranges) of the platinum concentrations are known in the prior art, a person having ordinary skill in the art would have had it within his or her ordinary capabilities through routine experimentation and optimization to arrive at platinum concentrations at the lower and upper ends of 505 that are within the respective claimed ranges as applicant has not disclosed that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical, and it appears prima facie that the process would possess utility using another concentration in each region. Moreover, it has been held that limitations directed to optimization of element amounts are prima facie obvious absent a disclosure that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical.

Further, it would have been obvious to one having ordinary skill in the art at the time the invention was made to modify the structure of Okuno in view of Dussarrat in view of Lee in view of Cabral (which teaches that a platinum concentration gradient, a result effective variable as the platinum amount in the gradient formed by Okuno using the ranges taught by Dussarrat in view of Lee in view of Cabral affects the electrical resistance and heat-resistant temperature of the nickel silicide) to form the nickel silicide containing platinum with a platinum gradient that falls within the claimed ranges unless it is shown that the claimed range achieves unexpected results relative to the prior art range. *In re Woodruff*, 16 USPQ2d 1935, 1937 (Fed. Cir. 1990). See also *In re Boesch*, 205 USPQ 215 (CCPA) (discovery of optimum value of result effective variable in known process is ordinarily within skill of art) and *In re Aller*, 105 USPQ 233 (CCPA 1955) (selection of optimum ranges within prior art general conditions is obvious).

11. Claims 18 and 19 are rejected under 35 U.S.C. 103(a) as being unpatentable over Okuno in view of Wang et al (US 2005/0095799 and Wang hereinafter).

As to claims 18 and 19: Although the structure disclosed by Okuno shows substantial features of the claimed invention (discussed in paragraph 7 above), it fails to expressly disclose:

wherein the nickel silicide layer has a thickness of about 100 Å to about 220 Å; wherein the nickel silicide layer has a thickness of about 100 Å to about 150 Å.

Okuno discloses a nickel silicide layer 502 in [0144], but fails to expressly disclose a thickness. Wang discloses in [0019] that the silicide for the gate and source/drain regions can be formed to between about 100 Angstroms to about 300 Angstroms. Therefore, the claimed range and the prior art ranges overlap. As stated in MPEP 2144.05, "[i]n the case where the claimed ranges 'overlap or lie inside ranges disclosed by the prior art' a *prima facie* case of obviousness exists." *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990).

In any case, it would have been ascertainable by routine experimentation and optimization to choose a silicide thickness within the particular claimed thickness range because applicant has not disclosed that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical, and it appears *prima facie* that the process would possess utility using another thickness. Moreover, it has been held that limitations directed to optimization of layer thickness are *prima facie* obvious absent a disclosure that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical.

12. Claim 21 is rejected under 35 U.S.C. 103(a) as being unpatentable over Okuno in view of Applicant's Admitted Prior Art (AAPA hereinafter).

Although the structure disclosed by Okuno shows substantial features of the claimed invention (discussed in paragraph 7 above), it fails to expressly disclose:

wherein the silicon substrate comprises a strained silicon germanium (Si-Ge) layer on a silicon layer.

Okuno discloses that the silicon substrate comprises only a silicon substrate. AAPA discloses that the use of a strained Si-Ge layer 101a on a Si layer 101b as the silicon substrate is well known in the art (see Fig. 1 and [0006]).

Therefore, given the teachings of AAPA, a person having ordinary skill in the art at the time of the invention would have readily recognized the desirability and advantages of modifying Okuno by employing the well known or conventional features of transistor fabrication, such as displayed by AAPA, by employing a silicon substrate comprising a strained Si-Ge layer on a Si layer in order to improve electrical characteristics of the transistors formed thereon ([0006]-[0008]).

13. Claims 22-24 are rejected under 35 U.S.C. 103(a) as being unpatentable over Okuno in view of Sun et al (US 2005/0247926 and Sun hereinafter).

As to claims 22-24: Although the structure disclosed by Okuno shows substantial features of the claimed invention (discussed in paragraph 7 above), it fails to expressly disclose:

a stress liner on the nickel silicide layer; an oxide or oxynitride layer on the stress liner; wherein the stress liner comprises stressed silicon nitride.

In the same field of endeavor, Sun discloses **a stress liner (90) on the nickel silicide layer** (Fig. 10; [0030]-[0031]; NiSi layer 76 and 77); **an oxide or oxynitride**

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layer (100) on the stress liner (Fig. 10; [0031]); wherein the stress liner comprises stressed silicon nitride ([0031]).

Given the teachings of Sun, a person having ordinary skill in the art at the time of the invention would have readily recognized the desirability and advantages of modifying Okuno by employing the well known or conventional features of transistor fabrication, such as displayed by Sun, by employing a stressed SiN layer over the transistor/NiSi layers and an oxide or oxynitride layer on the stress liner in order to improve the drive current of the device using the stressed layer and using the oxide layer as a layer to protect the stressed layer during subsequent processing ([0007]).

14. Claims 14, 22-27, 33, and 34 are rejected under 35 U.S.C. 103(a) as being unpatentable over Sun in view of Okuno.

As to claims 14, 22-27, 33, and 34: Sun discloses **a semiconductor device comprising: a silicon substrate** (Fig. 14; [0030]; substrate comprises 70 and 71), **a P-channel transistor (PMOS) and an N-channel transistor (NMOS), each comprising source/drain regions (source/drain regions) and a gate electrode (72), on the silicon substrate** (Fig. 14; [0030]); **and a first nickel silicide layer (76 and 77; 76 is on the gate while 77 is on the source/drain regions) on the gate electrode and source/drain regions of the P-channel transistor** (Fig. 14; [0030]); **and a second nickel silicide (76 and 77; 76 is on the gate while 77 is on the source/drain regions) layer on the gate electrode and source/drain regions of the N-channel transistor** (Fig. 14; [0030]); **a first stress liner (90) on the P-channel transistor/on the first nickel silicide layer**

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(Fig. 14; [0031]); **and a second stress liner (120) on the N-channel transistor/on the second nickel silicide layer** (Fig. 14; 0033); **wherein the first stress liner (90) exhibits high compressive stress and the second stress liner (120) exhibits high tensile stress** ([0031] and [0033]; 90 exhibits a high amount of compressive stress and 120 exhibits a high amount of tensile stress); **an oxide or oxynitride layer on the first and second stress liners** (Fig. 14; [0031 and [0033]; layer 100 is on first stress liner 90 and layer 130 is on second stress liner 120; 100 and 130 can be oxide or oxynitride); **wherein the first (90) and second (120) stress liners comprise stressed silicon nitride** ([0031] and [0033]; 90 and 120 can comprise silicon nitride).

Sun fails to expressly disclose **wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient with platinum increasing in amount in a direction away from the P-channel and N-channel transistors.**

In the same field of endeavor, Okuno discloses forming a nickel silicide layer on transistors in a high-heat resistance area (an area where CMOS transistors are formed), see Fig. 21 and [0081]. Okuno discloses forming a platinum region 505 within the nickel silicide 502 of the transistors (see Fig. 21 and [0144] and [0148]). Okuno also discloses that the platinum concentration increases from a bottom region of 505 toward a top surface thereof (see [0148] and [0130]).

Given the teachings of Okuno, a person having ordinary skill in the art at the time of the invention would have readily recognized the desirability and advantages of modifying Sun by employing the well known or conventional features of CMOS

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transistor fabrication, such as displayed by Okuno, by employing a platinum concentration gradient within a nickel silicide, where the concentration gradient increases from a bottom region toward a top region away from the substrate, in order to increase the heat-resistance of the nickel silicide layers ([0141]).

15. Claims 16, 17, 28, and 29 are rejected under 35 U.S.C. 103(a) as being unpatentable over Sun in view of Okuno as applied to claims 14 and 27 above, and further in view of Dussarrat in view of Lee in view of Cabral.

As to claims 16, 17, 28, and 29: Sun in view of Okuno in paragraph 14 discloses that the region 505 (platinum containing region) of the nickel silicide layer 502 has a gradient that changes (increases) from the bottom of 505 toward the top of 505.

Okuno fails to expressly disclose the concentration of platinum within the layer 505.

Therefore, Sun in view of Okuno fails to expressly disclose **wherein the concentration gradient of platinum in each nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %; wherein the concentration gradient of platinum in each nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.**

Dussarrat discloses that platinum doping in the range of 5-10 wt. % are used in nickel silicide formation (see [0006]). Lee discloses in the Abstract that a 5-33 wt. % for platinum in a nickel silicide for p-MOSFETs is well known. Cabral discloses that n-MOSFETs can also use nickel silicides containing platinum is well known (see [0073]).

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Thus, Lee in view of Cabral teaches that nickel silicides containing platinum can be applied to CMOS applications as both NMOS and PMOS devices can use the silicide layers. Therefore, platinum concentrations within both claimed ranges are used and well known in the prior art.

As Okuno discloses that the platinum concentration increases from a lower portion of the layer 505 toward the top surface, and platinum concentrations within both (the upper and lower ranges) of the platinum concentrations are known in the prior art, a person having ordinary skill in the art would have had it within his or her ordinary capabilities through routine experimentation and optimization to arrive at platinum concentrations at the lower and upper ends of 505 that are within the respective claimed ranges as applicant has not disclosed that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical, and it appears prima facie that the process would possess utility using another concentration in each region. Moreover, it has been held that limitations directed to optimization of element amounts are prima facie obvious absent a disclosure that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical.

Further, it would have been obvious to one having ordinary skill in the art at the time the invention was made to modify the structure of Sun in view of Okuno in view of Dussarrat in view of Lee in view of Cabral (which teaches that a platinum concentration gradient, a result effective variable as the platinum amount in the gradient formed by Sun in view of Okuno using the ranges taught by Dussarrat in view of Lee in view of Cabral affects the electrical resistance and heat-resistant temperature of the nickel

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silicide) to form the nickel silicide containing platinum with a platinum gradient that falls within the claimed ranges unless it is shown that the claimed range achieves unexpected results relative to the prior art range. *In re Woodruff*, 16 USPQ2d 1935, 1937 (Fed. Cir. 1990). See also *In re Boesch*, 205 USPQ 215 (CCPA) (discovery of optimum value of result effective variable in known process is ordinarily within skill of art) and *In re Aller*, 105 USPQ 233 (CCPA 1955) (selection of optimum ranges within prior art general conditions is obvious).

16. Claims 18, 19, 30, and 31 are rejected under 35 U.S.C. 103(a) as being unpatentable over Sun in view of Okuno as applied to claims 14 and 27 above, and further in view of Wang.

As to claims 18, 19, 30, and 31: Although the structure disclosed by Sun in view of Okuno shows substantial features of the claimed invention (discussed in paragraph 14 above), it fails to expressly disclose:

wherein each nickel silicide layer has a thickness of about 100 Å to about 220 Å; wherein each nickel silicide layer has a thickness of about 100 Å to about 150 Å.

Sun combined with Okuno discloses a nickel silicide layer 76 and 77 in [0030] of Sun, but Sun in view of Okuno fails to expressly disclose a thickness. Wang discloses in [0019] that the silicide for the gate and source/drain regions can be formed to between about 100 Angstroms to about 300 Angstroms. Therefore, the claimed range and the prior art ranges overlap. As stated in MPEP 2144.05, “[i]n the case where the claimed

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ranges “overlap or lie inside ranges disclosed by the prior art” a *prima facie* case of obviousness exists.” *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (COPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990).

In any case, it would have been ascertainable by routine experimentation and optimization to choose a silicide thickness within the particular claimed thickness range because applicant has not disclosed that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical, and it appears *prima facie* that the process would possess utility using another thickness. Moreover, it has been held that limitations directed to optimization of layer thickness are *prima facie* obvious absent a disclosure that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical.

17. Claims 21 and 32 are rejected under 35 U.S.C. 103(a) as being unpatentable over Sun in view of Okuno as applied to claims 14 and 27 above, and further in view of AAPA.

As to claims 21 and 32: Although the structure disclosed by Sun in view of Okuno shows substantial features of the claimed invention (discussed in paragraph 14 above), it fails to expressly disclose:

wherein the silicon substrate comprises a strained silicon germanium (Si-Ge) layer on a silicon layer.

Sun combined with Okuno discloses that the silicon substrate comprises a strained silicon layer overlying a Si-Ge layer. In [0004] of Sun, the construction is described as improving drive currents for transistors formed thereon.

AAPA discloses that the use of a strained Si-Ge layer 101a on a Si layer 101b as the silicon substrate is well known in the art (see Fig. 1 and [0006]).

The improvements to device performance (electron flow is 70% faster and operating speeds are 35% faster, compared to traditional Si) is identical for the constructions.

The two constructions, in view of the teachings of Sun and AAPA, are art recognized equivalent structures for CMOS substrates. Therefore, given the two constructions (a strained Si layer over a Si-Ge layer as taught by Sun and a strained Si-Ge layer over a Si layer as taught by AAPA) and that their benefits are substantially identical (given the disclosures), the two structures would be recognized as equivalent structures in the art. Therefore, because these two substrate structures are art recognized equivalents at the time the invention was made, one of ordinary skill in the art would have found it obvious to substitute the substrate structure of AAPA for the substrate structure of Sun in view of Okuno given that they are both taught to have the same benefits and the substitution would have yielded the predictable results of improving electrical characteristics of the transistors formed thereon ([0006]-[0008] of AAPA).

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18. Claim 35 is rejected under 35 U.S.C. 103(a) as being unpatentable over Sun in view of AAPA in view of Wang in view of Okuno in view of Dussarrat in view of Lee in view of Cabral.

Sun discloses **a semiconductor device comprising: a silicon substrate** (Fig. 14; [0030]; substrate comprises 70 and 71; the substrate is a silicon substrate as both portions comprise silicon), **a P-channel transistor (PMOS) and an N-channel transistor (NMOS), each comprising source/drain regions** (source/drain regions) **and a gate electrode (72), on the silicon substrate** (Fig. 14; [0030]); **a first nickel silicide layer** (76 and 77; 76 is on the gate while 77 is on the source/drain regions) **on the gate electrode and source/drain regions of the P-channel transistor** (Fig. 14; [0030]); **a second nickel silicide** (76 and 77; 76 is on the gate while 77 is on the source/drain regions) **layer on the gate electrode and source/drain regions of the N-channel transistor** (Fig. 14; [0030]); **a compressively stressed silicon nitride layer** (90) **on the first nickel silicide layer** (Fig. 14; [0031]); **and a tensile stressed silicon nitride layer** (120) **on the second nickel silicide layer** (Fig. 14; 0033]); **a first oxide or oxynitride layer** (100) **on the compressively stressed silicon nitride layer** (Fig. 14; [0031]; layer 100 is on first stress liner 90); **a second oxide or oxynitride layer** (90) **on the tensile stressed silicon nitride layer** (Fig. 14; [0033]; layer 90 is on second stress liner 120).

Sun fails to expressly disclose **a strained silicon germanium (Si-Ge) layer on the silicon substrate.**

Sun discloses that the silicon substrate comprises a strained silicon layer overlying a Si-Ge layer. In [0004] of Sun, the construction is described as improving drive currents for transistors formed thereon.

AAPA discloses that the use of a strained Si-Ge layer 101a on a Si layer 101b as the silicon substrate is well known in the art (see Fig. 1 and [0006]).

The improvements to device performance (electron flow is 70% faster and operating speeds are 35% faster, compared to traditional Si) is identical for the constructions.

The two constructions, in view of the teachings of Sun and AAPA, are art recognized equivalent structures for CMOS substrates. Therefore, given the two constructions (a strained Si layer over a Si-Ge layer as taught by Sun and a strained Si-Ge layer over a Si layer as taught by AAPA) and that their benefits are substantially identical (given the disclosures), the two structures would be recognized as equivalent structures in the art. Therefore, because these two substrate structures are art recognized equivalents at the time the invention was made, one of ordinary skill in the art would have found it obvious to substitute the substrate structure of AAPA for the substrate structure of Sun in view of Okuno given that they are both taught to have the same benefits and the substitution would have yielded the predictable results of improving electrical characteristics of the transistors formed thereon ([0006]-[0008] of AAPA).

Sun in view of AAPA fails to expressly disclose wherein each nickel silicide layer has **a thickness of about 100 Å to about 220 Å.**

Sun combined with AAPA discloses a nickel silicide layer 76 and 77 in [0030] of Sun, but Sun in view of AAPA fails to expressly disclose a thickness. Wang discloses in [0019] that the silicide for the gate and source/drain regions can be formed to between about 100 Angstroms to about 300 Angstroms. Therefore, the claimed range and the prior art ranges overlap. As stated in MPEP 2144.05, "[i]n the case where the claimed ranges 'overlap or lie inside ranges disclosed by the prior art' a *prima facie* case of obviousness exists." *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990).

In any case, it would have been ascertainable by routine experimentation and optimization to choose a silicide thickness within the particular claimed thickness range because applicant has not disclosed that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical, and it appears *prima facie* that the process would possess utility using another thickness. Moreover, it has been held that limitations directed to optimization of layer thickness are *prima facie* obvious absent a disclosure that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical.

Sun in view of AAPA in view of Wang fails to expressly disclose **wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient with platinum increasing in amount in a direction away from the P-channel and N-channel transistors.**

In the same field of endeavor, Okuno discloses forming a nickel silicide layer on transistors in a high-heat resistance area (an area where CMOS transistors are formed),

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see Fig. 21 and [0081]. Okuno discloses forming a platinum region 505 within the nickel silicide 502 of the transistors (see Fig. 21 and [0144] and [0148]). Okuno also discloses that the platinum concentration increases from a bottom region of 505 toward a top surface thereof (see [0148] and [0130]).

Given the teachings of Okuno, a person having ordinary skill in the art at the time of the invention would have readily recognized the desirability and advantages of modifying Sun in view of AAPA in view of Wang by employing the well known or conventional features of CMOS transistor fabrication, such as displayed by Okuno, by employing a platinum concentration gradient within a nickel silicide, where the concentration gradient increases from a bottom region toward a top region away from the substrate, in order to increase the heat-resistance of the nickel silicide layers ([0141]).

Sun in view of AAPA in view of Wang in view of Okuno fails to expressly disclose **wherein the concentration gradient of platinum in each nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %; wherein the concentration gradient of platinum in each nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.**

Sun in view of AAPA in view of Wang in view of Okuno idiscloses that the region 505 (platinum containing region) of the nickel silicide layer 502 has a gradient that changes (increases) from the bottom of 505 toward the top of 505 (see Okuno). Okuno fails to expressly disclose the concentration of platinum within the layer 505.

Dussarrat discloses that platinum doping in the range of 5-10 wt. % are used in nickel silicide formation (see [0006]). Lee discloses in the Abstract that a 5-33 wt. % for platinum in a nickel silicide for p-MOSFETs is well known. Cabral discloses that n-MOSFETs can also use nickel silicides containing platinum is well known (see [0073]). Thus, Lee in view of Cabral teaches that nickel silicides containing platinum can be applied to CMOS applications as both NMOS and PMOS devices can use the silicide layers. Therefore, platinum concentrations within both claimed ranges are used and well known in the prior art.

As Okuno discloses that the platinum concentration increases from a lower portion of the layer 505 toward the top surface, and platinum concentrations within both (the upper and lower ranges) of the platinum concentrations are known in the prior art, a person having ordinary skill in the art would have had it within his or her ordinary capabilities through routine experimentation and optimization to arrive at platinum concentrations at the lower and upper ends of 505 that are within the respective claimed ranges as applicant has not disclosed that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical, and it appears prima facie that the process would possess utility using another concentration in each region. Moreover, it has been held that limitations directed to optimization of element amounts are prima facie obvious absent a disclosure that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical.

Further, it would have been obvious to one having ordinary skill in the art at the time the invention was made to modify the structure of Sun in view of AAPA in view of

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Wang in view of Okuno in view of Dussarrat in view of Lee in view of Cabral (which teaches that a platinum concentration gradient, a result effective variable as the platinum amount in the gradient formed by Sun in view of AAPA in view of Wang in view of Okuno using the ranges taught by Dussarrat in view of Lee in view of Cabral affects the electrical resistance and heat-resistant temperature of the nickel silicide) to form the nickel silicide containing platinum with a platinum gradient that falls within the claimed ranges unless it is shown that the claimed range achieves unexpected results relative to the prior art range. *In re Woodruff*, 16 USPQ2d 1935, 1937 (Fed. Cir. 1990). See also *In re Boesch*, 205 USPQ 215 (CCPA) (discovery of optimum value of result effective variable in known process is ordinarily within skill of art) and *In re Aller*, 105 USPQ 233 (CCPA 1955) (selection of optimum ranges within prior art general conditions is obvious).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Joseph C. Nicely whose telephone number is (571)270-3834. The examiner can normally be reached on Monday through Friday 8:30AM-6:00PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Matthew Landau can be reached on (571) 272-1731. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/Joseph C. Nicely/
Examiner, Art Unit 2813

/Matthew C. Landau/
Supervisory Patent Examiner, Art Unit 2813

Notice of References Cited	Application/Control No. 13/096,511		Applicant(s)/Patent Under Reexamination RAMANI ET AL.	
	Examiner Joseph C. Nicely		Art Unit 2813	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-2005/0095799 A1	05-2005	Wang et al.	438/303
*	B	US-2007/0020929 A1	01-2007	Purtell et al.	438/682
*	C	US-2007/0034967 A1	02-2007	Nayfeh et al.	257/369
*	D	US-2007/0093047 A1	04-2007	Okuno et al.	438/592
*	E	US-2008/0090369 A1	04-2008	AKIYAMA et al.	438/308
*	F	US-2008/0237750 A1	10-2008	Tsai et al.	257/412
*	G	US-2008/0274611 A1	11-2008	Cabral et al.	438/653
*	H	US-2009/0028744 A1	01-2009	Hui et al.	420/456
*	I	US-2010/0256405 A1	10-2010	DUSSARRAT et al.	556/32
	J	US-			
	K	US-			
	L	US-			
	M	US-			

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*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
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	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	Lee et al. "Fully Silicided Ni1-xPtSi Metal Gate Electrode for p-MOSFETS", Electrochem. Solid-State Lett., Vol. 8, Issue 7, pp. G156-G159, May 12, 2005
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Receipt date: 04/28/2011

PTO/SB/08a (07-09)

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				Application Number	13/096,511
				Filing Date	04-28-11
				First Named Inventor	Karthik Ramani
				Art Unit	2813
				Examiner Name	J. Nicely
Sheet	1	of	1	Attorney Docket Number	P2343US01

[illegible][illegible]

Examiner Signature	/Joseph C. Nicely/	Date Considered	04/11/2012
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CONFIRMATION NO. 4709

SERIAL NUMBER 13/096,511	FILING or 371(c) DATE 04/28/2011 RULE	CLASS 257	GROUP ART UNIT 2813	ATTORNEY DOCKET NO. P2343US01		
APPLICANTS Karthik Ramani, Hopewell Junction, NY; Paul R. BESSER, Sunnyvale, CA; ** CONTINUING DATA ***** This application is a DIV of 12/366,378 02/05/2009 PAT 7,994,038 ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 05/10/2011						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/JOSEPH C NICELY/</u> Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY NY	SHEETS DRAWINGS 13	TOTAL CLAIMS 20	INDEPENDENT CLAIMS 3
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TITLE METHOD TO REDUCE MOL DAMAGE ON NiSi						
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☐ 5	NiSi AND platinum, (gradient OR concentration OR varying OR varies)	0 Metadata Apr. 11, 2012 16:12 UTC
☐ 4	NiSi AND platinum	76 Metadata Apr. 11, 2012 16:12 UTC
☐ 3	silicide AND platinum AND (concentration OR gradient) AND (changing OR varying OR varies OR increases OR decreases OR high OR low OR great OR less)	0 Metadata Apr. 11, 2012 16:12 UTC
☐ 2	silicide AND platinum AND (concentration OR gradient), changing OR varying OR varies OR increases\$5 OR decreases\$ OR high\$5 OR low\$4 OR great\$6 OR less\$5	0 Metadata Apr. 11, 2012 16:11 UTC
☐ 1	silicide AND platinum AND (concentration OR gradient)	2926 Metadata Apr. 11, 2012 16:11 UTC

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Search Notes 	Application/Control No. 13096511	Applicant(s)/Patent Under Reexamination RAMANI ET AL.
	Examiner JOSEPH C NICELY	Art Unit 2813

SEARCHED			
Class	Subclass	Date	Examiner
257	369, 383, 412, E29.139, E29.242	4/11/2012	JCN

SEARCH NOTES		
Search Notes	Date	Examiner
EAST Search (see attached history)	4/11/2012	JCN
Google Search (see attached history)	4/11/2012	JCN
IEEE Xplore Search (see attached history)	4/11/2012	JCN
Inventor Name Search and Assignee Search conducted through EAST	4/11/2012	JCN

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	2	("20050247926" "6841449").PN.	US-PGPUB; USPAT; FPRS; EPO; JPO; IBM_TDB	ADJ	ON	2012/04/11 10:29
S3	16	(gate electrode OR electrode) WITH (platinum OR Pt) NEAR4 (gradient OR varying OR varies) NEAR2 concentration	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 10:35
S4	6	("20050247926" "20060051961" "20060284263" "20070093047" "20090127594" "6841449").PN.	US-PGPUB; USPAT; USOCR	ADJ	ON	2012/04/11 10:36
S5	18	(gate electrode OR electrode) SAME (platinum OR Pt) NEAR4 (gradient OR varying OR varies) NEAR2 concentration	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 10:40
S6	20	(gate electrode OR electrode or silicide or salicide) SAME (platinum OR Pt) NEAR4 (gradient OR varying OR varies) NEAR2 concentration	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 10:40
S7	4	S6 not S3	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 10:40
S8	0	("7994038").URPN.	USPAT	ADJ	ON	2012/04/11 10:42
S9	2020	(platinum OR Pt) NEAR4 (gradient OR concentration) NEAR4 (changing OR varying OR varies OR increas\$5 OR decreas\$ OR high\$5 OR low\$4 OR great\$5 OR less\$5)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 10:43
S10	270	(electrode OR gate OR silicide) SAME (platinum OR Pt) NEAR4 (gradient OR concentration) NEAR4 (changing OR varying OR varies OR increas\$5 OR decreas\$ OR high\$5 OR low\$4 OR great\$5 OR less\$5)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 10:43
S11	2	(12/366378).APP.	US-PGPUB; USPAT; USOCR	OR	OFF	2012/04/11 11:03

S12	1	"11942094"	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 11:09
S13	2	("2009/0127594").URPN.	USPAT	ADJ	ON	2012/04/11 11:09
S14	1	(11/942094).APP.	US-PGPUB; USPAT; USOCR	OR	OFF	2012/04/11 11:10
S15	4	("20070004205" "20090028744" "20070093047" "20070093047" "20070004205" "20090127594").PN.	US-PGPUB; USPAT; FPRS; EPO; JPO; IBM_TDB	ADJ	ON	2012/04/11 11:15
S16	5	("20070238321" "20080090369" "20080093666" "20080242035" "6218249").PN.	US-PGPUB; USPAT; USOCR	ADJ	ON	2012/04/11 11:35
S17	9	("20030094662" "20050101076" "20060038229" "20060292799" "20070052039" "20070215956" "20070224808" "20100193876" "6987050").PN.	US-PGPUB; USPAT; USOCR	ADJ	ON	2012/04/11 11:38
S18	17	("20040201066" "20050009328" "20050148114" "20060017112" "20060043497" "20060163661" "20060194393" "20080029822" "6030863" "6365446" "6475874" "6528981" "6905922" "7067379" "7078278" "7141467" "7151023").PN.	US-PGPUB; USPAT; USOCR	ADJ	ON	2012/04/11 11:41
S19	58	(NiPt OR silicide) SAME (platinum OR Pt) NEAR4 (gradient OR concentration) NEAR4 (changing OR varying OR varies OR increas\$5 OR decreas\$ OR high\$5 OR low\$4 OR great\$5 OR less\$5)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 11:51
S20	3	S19 not (S3 or S6 or S10)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 11:52
S21	3	(salicide) SAME (platinum OR Pt) NEAR4 (gradient OR concentration) NEAR4 (changing OR varying OR varies OR increas\$5 OR decreas\$ OR high\$5 OR low\$4 OR great\$5 OR less\$5)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 11:52
S22	1	(13/096511).APP.	US-PGPUB; USPAT; USOCR	OR	OFF	2012/04/11 11:53
S23	16	(nickel silicide OR NiSi) SAME (platinum OR Pt) NEAR4 (gradient OR concentration) NEAR4 (changing OR varying OR varies OR increas\$5 OR decreas\$ OR high\$5 OR low\$4 OR great\$5 OR less\$5)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 11:53
S24	3832	257/369.ccls.	US-PGPUB; USPAT;	ADJ	ON	2012/04/11 12:00

			FPRS; EPO; JPO; DERWENT; IBM_TDB			
S25	3	S24 AND (platinum OR Pt) NEAR4 (gradient OR concentration) NEAR4 (changing OR varying OR varies OR increas\$5 OR decreas\$ OR high\$5 OR low\$4 OR great\$5 OR less\$5)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:00
S26	423	257/383.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:01
S27	2	S26 AND (platinum OR Pt) NEAR4 (gradient OR concentration) NEAR4 (changing OR varying OR varies OR increas\$5 OR decreas\$ OR high\$5 OR low\$4 OR great\$5 OR less\$5)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:01
S28	1795	257/412.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:01
S29	5	S28 AND (platinum OR Pt) NEAR4 (gradient OR concentration) NEAR4 (changing OR varying OR varies OR increas\$5 OR decreas\$ OR high\$5 OR low\$4 OR great\$5 OR less\$5)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:01
S30	0	(S24 or S26 or S28) AND (platinum OR Pt) Near4 (gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:02
S31	19	(NiPt OR silicide) SAME (platinum OR Pt) Near4 (gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:02
S32	116	257/E29.139.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:03
S33	0	(S32) AND (platinum OR Pt) Near4 (gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:03
S34	0	S32 AND (platinum OR Pt) NEAR4 (gradient OR concentration) NEAR4	US-PGPUB; USPAT;	ADJ	ON	2012/04/11 12:04

		(changing OR varying OR varies OR increas\$5 OR decreas\$ OR high\$5 OR low\$4 OR great\$5 OR less\$5)	FPRS; EPO; JPO; DERWENT; IBM_TDB			
S35	676	257/E29.242.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:04
S36	1	(S35) AND (platinum OR Pt) Near4 (gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:04
S37	2	S35 AND (platinum OR Pt) NEAR4 (gradient OR concentration) NEAR4 (changing OR varying OR varies OR increas\$5 OR decreas\$ OR high\$5 OR low\$4 OR great\$5 OR less\$5)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:04
S38	47	(karthik near ramani).in.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:05
S39	3	(S38) AND (platinum OR Pt) Near4 (gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:05
S40	3	S38 AND (platinum OR Pt) NEAR4 (gradient OR concentration) NEAR4 (changing OR varying OR varies OR increas\$5 OR decreas\$ OR high\$5 OR low\$4 OR great\$5 OR less\$5)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:05
S41	0	(paul besser).in.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:06
S42	262	(paul near besser).in.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:06
S43	3	S42 AND (platinum OR Pt) Near4 (gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:06
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S50	0	S49 not S48	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 12:08
S51	59	(nickel silicide OR NiSi) WITH ("1"\$2 near2 angstrom OR "2"\$2 near2 angstrom)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/11 15:08
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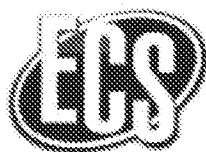
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<i>Index of Claims</i> 	Application/Control No. 13096511	Applicant(s)/Patent Under Reexamination RAMANI ET AL.
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Fully Silicided $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ Metal Gate Electrode for p-MOSFETs

R. T. P. Lee, S. L. Liew, W. D. Wang, E. K. C. Chua, S. Y. Chow,
M. Y. Lai, and D. Z. Chi*

Institute of Materials Research and Engineering, Singapore 117602

We investigated $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ ($x = 0.05-0.33$) as a metal gate electrode for p-metal oxide semiconductor field effect transistors (MOSFETs). Our results showed that, with a proper atomic composition of $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$, the work function of the gate electrode can be tuned from ~ 4.59 eV (for NiSi) to ~ 5.21 eV (for $\text{Ni}_{0.67}\text{Pt}_{0.33}\text{Si}$). Negligible variations in both the extracted flatband voltages and gate oxide thicknesses with processing temperatures (up to 900°C) demonstrated the excellent thermal stability of the $\text{Ni}_{0.67}\text{Pt}_{0.33}\text{Si-SiO}_2$ gate stacks. High-resolution transmission electron microscopy also showed no evidence of degradation at the $\text{Ni}_{0.67}\text{Pt}_{0.33}\text{Si-SiO}_2$ interface and the formation of Ni and/or Pt precipitates at or beneath the $\text{SiO}_2\text{-Si}$ interface.
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Manuscript submitted February 2, 2005; revised manuscript received March 29, 2005. Available electronically May 12, 2005.

The introduction of metal gate electrodes in nanoscale complementary metal-oxide-semiconductor (CMOS) devices is imperative to overcome problems associated with poly-Si gate such as poly-depletion effects and boron penetration.¹ However, to achieve optimal operation of high performance bulk CMOS devices, the required gate work functions for n- and p-channel transistors should be within 0.2 eV of the conduction and valence bandedges of Si, respectively.² While tailoring the work functions of poly-Si is achieved readily by dopant incorporation, this may not be true for a metal gate electrode; hence the achievement of appropriate metal gate work functions for nanoscale CMOS devices is crucial to realizing dual metal gate CMOS devices. Recently, fully silicided NiSi was shown to be an attractive alternative to elemental metals for metal gate electrode applications.³⁻⁸ It was demonstrated that fully silicided NiSi gates effectively reduce poly-depletion effects and have lower sheet resistance with a work function that can be tuned by the implantation of different dopants (e.g., As, P, B, and BF_3) at varying implant doses into a silicon film before full silicidation. Cabral and co-workers have also recently shown that the work function of the fully silicided NiSi gate can be tuned by the full silicidation of poly-Si or Al-doped poly-Si by a Ni(Pt) alloy, increasing the work function value of the fully silicided NiSi gate to 4.76 and 4.96 eV, respectively.⁹ In this letter, we investigated the dependence of Ni(Pt)Si gate work function on the Pt content and demonstrated that a continuous and tunable work function from ~ 4.59 to ~ 5.21 eV could be achieved. The effects of Pt incorporation on the Ni(Pt)Si-SiO₂ and SiO₂-Si interface integrity were also examined, which revealed no evidence of degradation. Negligible variations in the extracted flatband voltages and gate oxide thickness from 400 to 900°C rapid thermal annealing also demonstrate the excellent thermal stability of the $\text{Ni}_{0.67}\text{Pt}_{0.33}\text{Si-SiO}_2$ interface.

$\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ -gated MOS capacitors with different Pt content ($x = 0.05, 0.10, 0.25$, and 0.33 , calibrated by Rutherford backscattering for $x = 0.05$ and 0.10 with $x = 0.25$ and 0.33 estimated by the ratios of the original Ni and Pt layer thicknesses and atomic densities) were fabricated on (100) p-type silicon substrates with thermally grown SiO₂ as the gate oxide. In order to investigate the work function dependence of $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ gates with varying Pt content, SiO₂ gate oxides with various thicknesses (22, 44, 58, and 82 nm) were obtained by etching down a 100 nm thick thermally grown SiO₂ gate oxide on the same Si wafer in buffered HF solution. SiO₂ gate oxide of ~ 3.6 nm in thickness was also grown by rapid thermal oxidation to investigate the effects of Ni(Pt)Si on the structural integrity of thin gate oxides. The circular-shaped Ni(Pt)Si-gated MOS capacitors with an area of $7.85 \times 10^{-3} \text{ cm}^2$ were formed by depositing undoped amorphous Si (30 nm) through a shadow mask on the SiO₂ gate oxide by radio frequency (rf) magnetron sputtering. 40 nm Ni, $\text{Ni}_{0.05}\text{Pt}_{0.95}$, and $\text{Ni}_{0.67}\text{Pt}_{0.33}$ single-layer films as well as

Pt(12 nm)/Ni(28 nm) and Pt(16 nm)/Ni(24 nm) bilayer films were then deposited onto the undoped amorphous Si by dc magnetron sputtering. We have used the bilayer scheme to fabricate $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ -gated MOS capacitors with Pt contents ($x = 0.25$ and 0.33) since no Ni(Pt) alloy targets with such high Pt content were available for our present study. The single-layer films were sputter-deposited from a Ni(Pt) alloy target while the bilayer films were sputter-deposited from Ni and Pt targets in a system with a base pressure of $< 3 \times 10^{-7}$ Torr. (Note: since NiSi and PtSi are completely miscible in each other, a full silicidation reaction of Pt-Ni bilayer film with Si results in the formation of a $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ solid solution.¹⁰) The $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ -gated MOS capacitors were then subjected to rapid thermal annealing (RTA) at temperatures from 400 to 900°C in N₂ ambient for 20 s, followed by the removal of any unreacted Pt and Ni films using a dilute aqua regia ($\text{HCl}:\text{HNO}_3:\text{H}_2\text{O} = 3:1:2$) and SPM solution ($\text{H}_2\text{O}_2:\text{H}_2\text{SO}_4 = 1:1$), respectively. Finally, a 200 nm thick Au film was deposited on the back side of the capacitors (after removal of the native oxide using dilute HF) to form ohmic contacts for electrical measurements. High-frequency capacitance-voltage (C-V) measurements were carried out on the $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ -gated MOS capacitors to examine the work function modification by varying the Pt content. Secondary ion mass spectroscopy (SIMS) was employed to obtain redistribution profiles of Ni, Si, Pt, and O in the $\text{Ni}_{1-x}\text{Pt}_x\text{Si-SiO}_2$ gate stack after different temperature RTA treatments. High resolution cross-sectional transmission electron microscopy (XTEM) was used to examine the quality of the $\text{Ni}_{1-x}\text{Pt}_x\text{Si-SiO}_2$ and SiO₂-Si interface and the gate oxide integrity after different temperature RTA treatments.

To obtain effective gate work function values, high frequency C-V curves were first obtained on all fabricated $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ -gated MOS capacitors. Figure 1 shows typical high frequency C-V curves of $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ -gated MOS capacitors formed by RTA at 400°C on 82.88 nm thick thermally grown SiO₂ gate dielectrics on the same p-Si substrate. It is evident that an increasing Pt content induces a parallel shift of C-V curves toward the positive direction of the x axis, which should be due to the increase in the flatband voltage (V_{FB}) values.⁸

V_{FB} values were extracted from the measured high frequency C-V curves and plotted against different gate oxide thickness (T_{ox}) for varying Pt content as shown in Fig. 2a. The effective work function values of $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ metal gates and fixed charge densities were then extracted by linearly extrapolating the $V_{\text{FB}}\text{-}T_{\text{ox}}$ plots to $T_{\text{ox}} = 0$ and using the relationships

*It should be noted that metal work function values obtained in this way should be considered as fair approximations after accounting for uncertainties in the determination of V_{FB} and the contributions of interface trap states.

* E-mail: dz.chi@imre.a-star.edu.sg

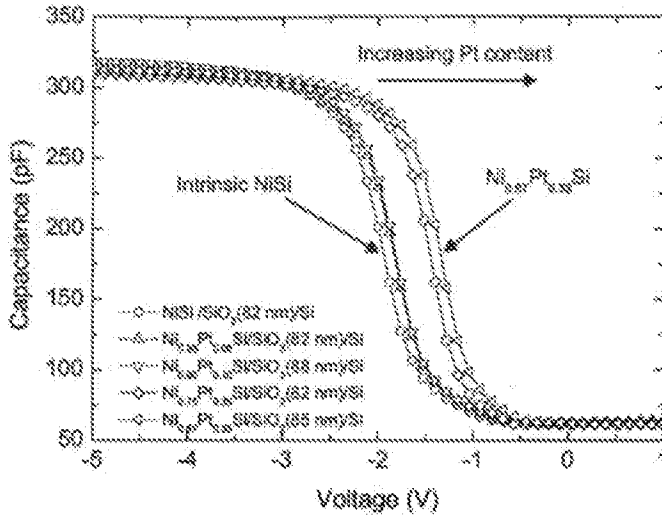


Figure 1. Effect of varying Pt content on the C-V characteristics of $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ alloy gate stacks formed by full silicidation of 30 nm thick undoped amorphous Si on 52–88 nm thick thermally grown SiO_2 gate dielectrics by RTA at 400°C for 20 s. Capacitor area is $7.85 \times 10^{-3} \text{ cm}^2$.

$$\Phi_{\text{MS}} = \Phi_{\text{M}} - \Phi_{\text{S}} \quad (1)$$

and

$$V_{\text{FB}} = \Phi_{\text{MS}} - \frac{Q_f}{\epsilon \epsilon_0} \quad (2)$$

where Φ_{MS} is the difference between metal (gate) and semiconductor (substrate) work functions, i.e., $\Phi_{\text{M}} - \Phi_{\text{S}}$, Q_f is the effective area density of fixed oxide charge, ϵ is dielectric constant of gate dielectrics (i.e., silicon oxide in this case), and ϵ_0 is the permittivity in vacuum. The results of the above analysis are summarized in the insets of Fig. 2a. Comparing the work function values of NiSi and $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ -gated MOS capacitors, it is evident that Pt incorporation affects the work function of NiSi gates strongly, shifting the work function of NiSi (4.59 eV for pure NiSi which is in good agreement with previously reported values¹) toward higher values with increasing Pt content. By varying the Pt content from 0.05 to 0.33, the work function varies almost linearly from ~ 4.75 to ~ 5.21 eV after 400°C, 20 s RTA as shown in Fig. 2b.

The larger work function values observed in $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ metal gates are certainly due to the fact that PtSi has a larger work function value than NiSi. PtSi shows a substantially higher Schottky barrier height on n-type Si as compared to NiSi, thus implying a larger work function value for PtSi.¹¹ It is known that NiSi and PtSi are completely miscible with each other and therefore able to form a solid solution in the form of $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ with x varying from 0 to 1. For a metal alloy of A_xB_{1-x} , the work function can be approximated by

$$\Phi_{\text{MAB}} = x\Phi_{\text{MA}} + (1-x)\Phi_{\text{MB}} + x(1-x)[(\Phi_{\text{MA}} - \Phi_{\text{MB}})(\rho_{\text{A}} - \rho_{\text{B}})/(\rho_{\text{A}} + (1-x)\rho_{\text{B}})] \quad (3)$$

where Φ_{MA} and Φ_{MB} are the pure constituent work functions of A and B, respectively, and ρ_{A} and ρ_{B} are the constituent total densities of states with $\rho_{\text{A}}/\rho_{\text{B}}$ being proportional to the ratio of the Fermi-level state densities.^{12–14} In a dilute A limit and/or when $\rho_{\text{A}}/\rho_{\text{B}} \sim 1$ as in the case of Ref. 13 and 14 [note: NiSi and PtSi are completely miscible in each with both NiSi and PtSi having the same orthorhombic structure (*Pnma* space group)], Eq. 3 reduces to

$$\Phi_{\text{MAB}} \approx \Phi_{\text{MB}} + x(\Phi_{\text{MA}} - \Phi_{\text{MB}})\rho_{\text{A}}/\rho_{\text{B}} \quad (4)$$

For a $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ alloy with $x < 0.5$ it is therefore reasonable to expect its work function to increase monotonically with increasing

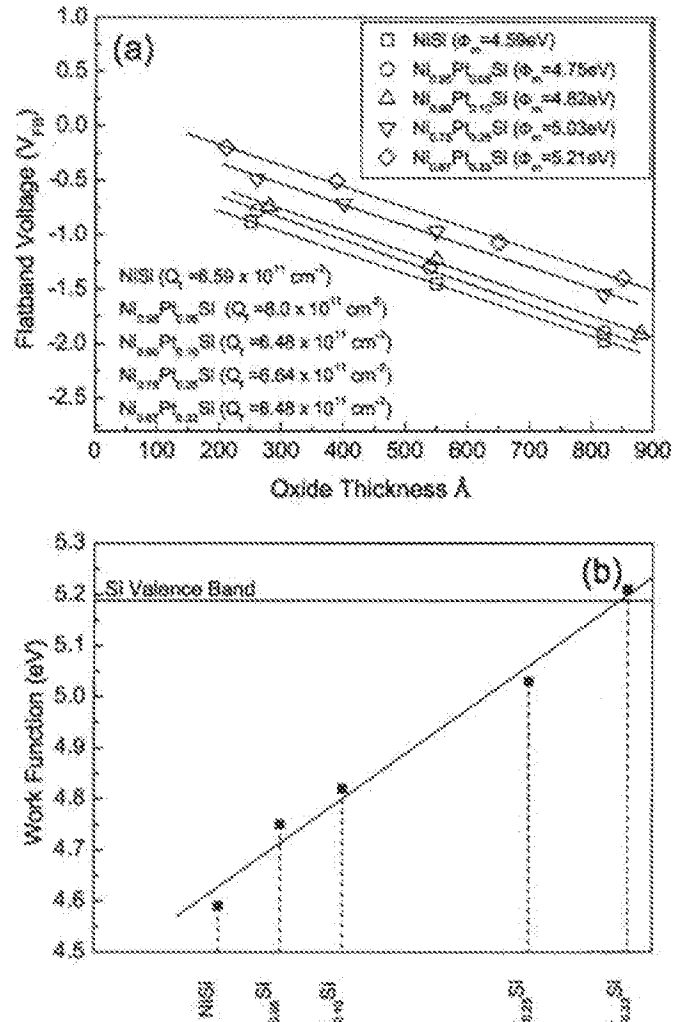


Figure 2. (a) Flatband voltage extrapolation as a function of gate oxide thickness with fully silicided $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ alloy gate electrodes. The insets list the Q_f and $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ work function values extracted. (b) Extracted $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ alloy work function as a function of Pt content.

Pt content, which concurs with the experimental result shown in Fig. 2b.

It is important to note that there is no noticeable change in the fixed oxide charge density (see Fig. 2a) after the incorporation of Pt. The observation of almost identical C-V curves (see Fig. 1) without voltage stretch-out distortions for all measured $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ -gated MOS capacitors also suggests minimal change in interfacial electronic state density. These negligible variations in fixed oxide charge density and interfacial electronic state density indicate that one can achieve a fully silicided NiSi gate with a Fermi level that matches to the valence bandedge of Si by using a Ni(Pt) alloy (e.g., $\text{Ni}_{1-x}\text{Pt}_x$ with $x \sim 0.25 - 0.33$) without degrading the gate dielectrics integrity, thus making $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ a suitable metal gate material for optimal p-MOSFET operation. It must be pointed out that other research groups have reported tunable NiSi work functions (achieved using various methods) for p-MOSFETs applications with work function values varying from ~ 4.76 eV⁵ to ~ 5.1 eV.⁶ Figure 3 summarizes and compares these reported NiSi work functions and the work functions obtained in this study.

The thermal stability of $\text{Ni}_{0.67}\text{Pt}_{0.33}\text{Si}$ alloy in a $\text{Ni}_{0.67}\text{Pt}_{0.33}\text{Si}/\text{SiO}_2$ gate stack was also investigated. Figure 4 shows the SIMS elemental depth profiles of $\text{Ni}_{0.67}\text{Pt}_{0.33}\text{Si}/\text{SiO}_2$ gate stack which was

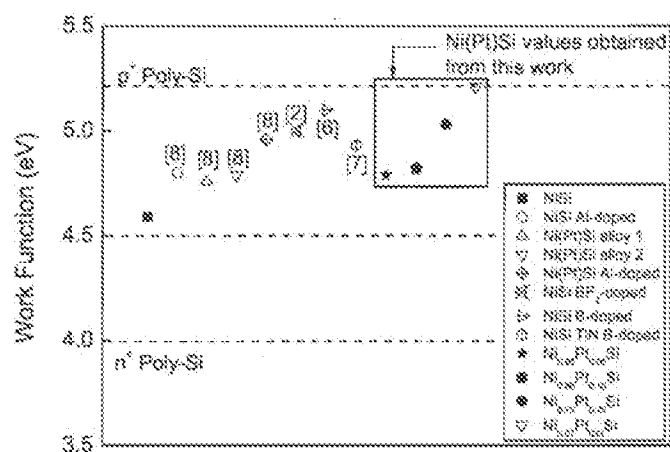


Figure 3. Summary of tunable NiSi work function values for p-MOSFETs plotted on a scale ranging from the position of the Fermi level to the valence band of silicon. The work function values of $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ alloy obtained in the present work are also plotted on the same scale for comparison.

RTA-silicided using Ni(24 nm)/Pt(16 nm) bilayer film at 900°C. It can be seen that the entire amorphous Si film was consumed with a uniform Pt distribution in the $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ bulk region. Due to the uniform distribution of Pt in the $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ layer, it is believed that the atomic composition of Pt in the entire $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ layer formed is close to 0.33, an average value for the entire $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ layer, with a slightly lower value $x = 0.31$ near the $\text{Ni}_{1-x}\text{Pt}_x\text{Si}/\text{SiO}_2$ interface (estimated from Ni/Pt signal ratio and taking 0.33 as the average x value). While the SIMS results clearly show that a fully silicided $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ gate with a uniform Pt distribution can be formed using Ni/Pt bilayer film, XTEM further reveals the formation of a uniformly thick $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ layer with a thin layer of excess Ni and/or Pt detected on the surface (see Fig. 5a). More importantly, the XTEM results (see Fig. 5b) show no evidence of degradation in the SiO_2 gate oxide (~ 3.6 nm) integrity and/or the formation of Ni and/or Pt precipitates at/or beneath the SiO_2 -Si interface even after a full silicidation at 900°C. This high structural and chemical stability of the $\text{Ni}_{1-x}\text{Pt}_x\text{Si}/\text{SiO}_2$ gate stacks observed is responsible for the high-temperature stability of the extracted V_{FB} and T_{ox} values (see Fig. 6) in these gate stacks. However, it is important to note that a slight increase in SiO_2 gate oxide thickness is observed in Fig. 5b for

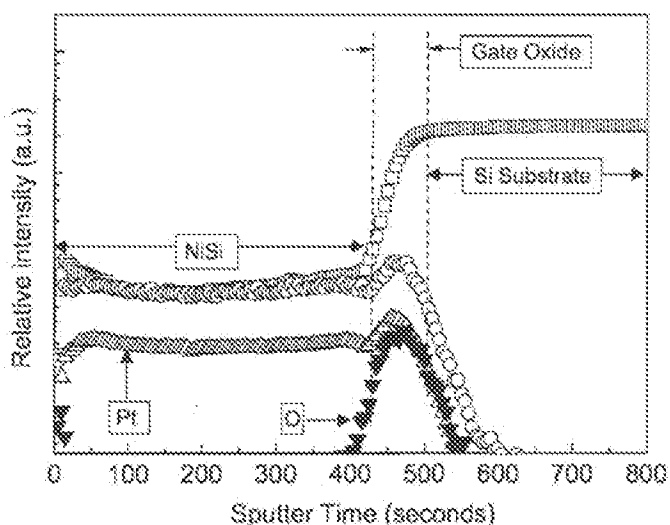


Figure 4. SIMS depth profiles of Ni, Pt, Si, and O after full silicidation at 900°C for 20 s in N_2 ambient.

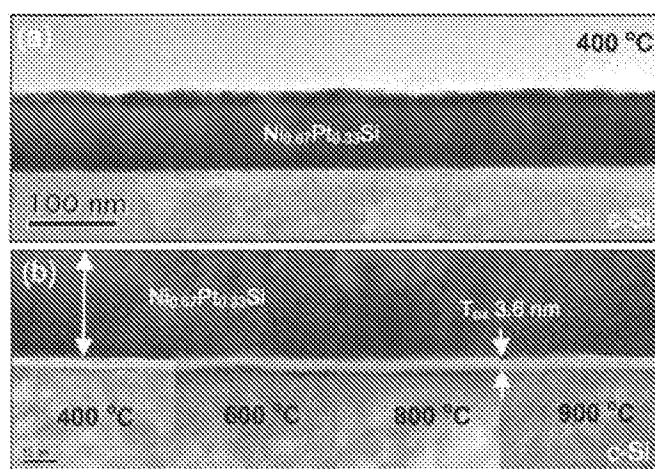


Figure 5. (a) XTEM of fully silicided $\text{Ni}_{0.6}\text{Pt}_{0.4}\text{Si}/\text{SiO}_2$ gate stack formed after RTA at 400°C for 20 s. (b) High-resolution XTEM of fully silicided $\text{Ni}_{0.6}\text{Pt}_{0.4}\text{Si}/\text{SiO}_2$ gate stacks formed at different RTA temperatures for 20 s.

temperatures $\geq 600^\circ\text{C}$. It is believed that this slight increase in gate oxide thickness is due to the oxidation reaction of the residual oxygen, which was incorporated into the sputter-deposited amorphous Si and Ni(Pt) layers, with Si at the amorphous Si- SiO_2 interface during silicidation reaction at high temperatures. The possibility of the reaction between Ni (and/or Pt) and SiO_2 could be ruled out as the reaction between Ni (and/or Pt) and SiO_2 is not favored thermodynamically.^{13,16} Though no structural defects were observed under TEM with a 3.6 nm SiO_2 gate oxide, the impact of the fully silicided $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ gate on ultrathin gate oxide integrity (GOI) remains for further evaluation, and we are currently in the process of addressing this concern.

We have demonstrated that the work function of NiSi can be tuned by Pt incorporation into the Ni film before silicidation, which results in the formation of a solid solution of Ni(Pt)Si alloy upon full silicidation. With a proper atomic composition of $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$, the work function of the gate electrode can be tuned continuously over a wide range, to meet the gate work function requirements for p-MOSFETs applications. No evidence of degradation in the integrity of the SiO_2 gate oxide (~ 3.6 nm) and formation of Ni and/or Pt precipitates at/or beneath the SiO_2 -Si interface was observed for RTA temperatures up to 900°C. Negligible variations in the extracted flatband voltages and effective gate oxide thickness also li-

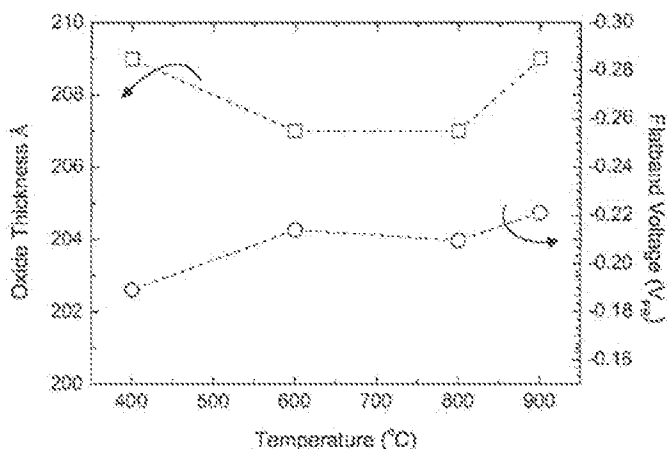


Figure 6. Variation of flatband voltages and effective gate oxide thickness as a function of RTA temperature in $\text{Ni}_{0.6}\text{Pt}_{0.4}\text{Si}/\text{SiO}_2$ gate stacks. The original oxide thickness in the $\text{Ni}_{0.6}\text{Pt}_{0.4}\text{Si}/\text{SiO}_2$ gate stacks was about 20 nm.

illustrate the excellent thermal stability of $\text{Ni}_{1-x}\text{Pt}_x\text{Si-SiO}_2$ gate stacks. Therefore, fully silicided $\text{Ni}_{1-x}\text{Pt}_x\text{Si}$ is a promising metal gate candidate for p-MOSFETs in a conventional dual-gate Si CMOS process.

Acknowledgments

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In re Application of:	Karthik Ramani <i>et al.</i>	Confirmation No.:	4709
Application No.:	13/096,511	Examiner:	Joseph C. Nicely
Filed:	April 28, 2011	Group Art Unit:	2813

For: METHOD TO REDUCE MOL DAMAGE ON NiSi

Commissioner for Patents
Alexandria, VA 22313-1450

RESPONSE UNDER 37 C.F.R. § 1.111

Dear Commissioner:

In response to the Office Action dated April 26, 2012, please amend this application as follows.

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AMENDMENT TO THE SPECIFICATION

Please amend the specification by marked up replacement paragraph(s) as follows.

[0006] Substrates based on “strained silicon” have attracted interest as a semiconductor material which provides increased speeds of electron and hole flow therethrough, thereby permitting fabrication of semiconductor devices with higher operating speeds, enhanced performance characteristics, and lower power consumption. A layer 101a of silicon-germanium (Si-Ge) is formed on a suitable crystalline substrate, e.g., a Si wafer or a silicon-on-insulator (SOI) wafer 101b. The Si-Ge layer has a greater lattice constant (spacing of Si and Ge atoms therein) relative to the underlying Si. As a consequence, the Si-Ge layer has a compressive ~~strain~~. strain. Electrons in such compressively strained layers have greater mobility than in conventional Si layers with smaller inter-atom spacings, i.e., there is less resistance to electron flow. For example, electron flow may be up to about 70% faster compared to electron flow in conventional Si. Transistors and IC devices formed with such strained Si layers can exhibit operating speeds up to about 35% faster than those of equivalent devices formed with conventional Si, without necessity for reduction in transistor size. Conventional practices based on strained silicon technology also involve epitaxially growing a relaxed silicon layer on a tensilely stressed silicon layer which is subsequently doped to form relaxed source/drain regions in the relaxed silicon layer.

[0042] Another embodiment of the present disclosure is schematically illustrated in FIGS. 7 through 15. Adverting to FIG. 7, a CMOS device is schematically illustrated comprising an NMOS transistor portion at the left and a PMOS transistor portion on the right, wherein similar features are denoted by similar reference characters. Strained Si-Ge layer 701 is formed over Si-~~Ge~~ layer 703. As in previous discussed embodiments, Si-Ge layer 701 may be globally strained or locally strained in the source/drain regions. Gate electrodes 705 are formed over strained Si-Ge layer 701 with gate dielectric 707 therebetween. Lightly doped source/drain regions 713a

are formed in Si-Ge layer 701 using gate electrodes 705 as a mask in a conventional manner. Sidewall spacers comprising oxide liner 709, as at a thickness of about 60 Å to about 600 Å, are formed on side surfaces of gate electrodes 705 and on portions of the upper surface of strained Si-Ge layer 701. Silicon oxide liner 709 can be formed in the same manner as discussed with respect to silicon oxide liner 209 in FIG. 2. Silicon nitride spacers 711 are then formed on silicon oxide liners 709 as by employing PECVD followed by etching.

[0043] Subsequently, heavily doped source/drain regions 713b are formed in Si-Ge layer 701 using gate electrodes 705 and sidewall spacers 709 and 711 as a mask, employing conventional implantation techniques, as by ion implanting boron, phosphorus, or arsenic.

[0049] A highly compressive stressed silicon nitride film 1101, ~~e.g.,~~ e.g., having a compressive stress greater than 1.5 GPa, is then deposited over both the NMOS and PMOS transistors as shown in FIG. 11. Deposition of highly compressive stressed silicon nitride film 1101 may be implemented in a conventional manner. Multiple layer deposition and plasma treatment further increases compressive stress. Subsequently, a thin oxide or oxynitride film 1103 is deposited, as illustrated in FIG. 11, by a conventional CVD process. Typically oxide or oxynitride film 1103 is deposited at a thickness of about 30 Å to about 60 Å.

AMENDMENT AND PRESENTATION OF CLAIMS

Please replace all prior claims in the present application with the following claims.

1. - 13. (Canceled)

14. (Currently Amended) A semiconductor device comprising:

a silicon substrate,

a transistor, comprising source/drain regions and a gate electrode, on the substrate; and

a nickel silicide layer on the gate electrode and source/drain regions,

wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt. % and about 12 wt. % at the silicon interface.

15. (Canceled)

16. (Original) The semiconductor device according to claim 14, wherein the concentration gradient of platinum in the nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

17. (Original) The semiconductor device according to claim 16, wherein the concentration gradient of platinum in the nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

18. (Original) The semiconductor device according to claim 14, wherein the nickel silicide layer has a thickness of about 100 Å to about 220 Å.

19. (Original) The semiconductor device according to claim 14, wherein the nickel silicide layer has a thickness of about 100 Å to about 150 Å.

20. (Canceled)

21. (Previously Presented) The semiconductor device according to claim 14, wherein the silicon substrate comprises a strained silicon germanium (Si-Ge) layer on a silicon layer.

22. (Previously Presented) The semiconductor device according to claim 14, further comprising a stress liner on the nickel silicide layer.

23. (Previously Presented) The semiconductor device according to claim 22, further comprising an oxide or oxynitride layer on the stress liner.

24. (Previously Presented) The semiconductor device according to claim 23, wherein the stress liner comprises stressed silicon nitride.

25. (Currently Amended) A semiconductor device comprising:

a silicon ~~substrate~~, substrate;

a P-channel transistor and an N-channel transistor, each comprising source/drain regions and

a gate electrode, on the silicon substrate; and

a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor; and

a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor,

wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the P-channel and N-channel transistors, from between about 8 wt. % and about 12 wt. % at the silicon interface.

26. (Previously Presented) The semiconductor device according to claim 25, further comprising:

a first stress liner on the P-channel transistor; and

a second stress liner on the N-channel transistor.

27. (Previously Presented) The semiconductor device according to claim 26, wherein the first stress liner exhibits high compressive stress and the second stress liner exhibits high tensile stress.

28. (Previously Presented) The semiconductor device according to claim 27, wherein the concentration gradient of platinum in each nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

29. (Previously Presented) The semiconductor device according to claim 28, wherein the concentration gradient of platinum in each nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

30. (Previously Presented) The semiconductor device according to claim 27, wherein each nickel silicide layer has a thickness of about 100 Å to about 220 Å.

31. (Previously Presented) The semiconductor device according to claim 30, wherein each nickel silicide layer has a thickness of about 100 Å to about 150 Å.

32. (Previously Presented) The semiconductor device according to claim 27, wherein the silicon substrate comprises a strained silicon germanium (Si-Ge) layer on a silicon layer.

33. (Previously Presented) The semiconductor device according to claim 27, further comprising an oxide or oxynitride layer on the first and second stress liners.

34. (Previously Presented) The semiconductor device according to claim 33, wherein the first and second stress liners comprise stressed silicon nitride.

35. (Currently Amended) A semiconductor device comprising:

a silicon substrate;

a strained silicon germanium (Si-Ge) layer on the silicon substrate;

a P-channel transistor comprising source/drain regions and a gate ~~electrode~~, electrode on the strained Si-Ge layer;

an N-channel transistor comprising source/drain regions and a gate ~~electrode~~, electrode on the strained Si-Ge layer, spaced from the P-channel transistor;

a first nickel silicide layer, having a thickness of about 100 Å to about 220 Å, on the gate electrode and source/drain regions of the P-channel transistor;

a second nickel silicide layer, having a thickness of about 100 Å to about 220 Å, on the gate electrode and source/drain regions of the N-channel transistor;

a compressively stressed silicon nitride layer on the first nickel silicide layer;

a tensile stressed silicon nitride layer on the second nickel silicide layer;

a first oxide or oxynitride layer on the compressively stressed silicon nitride layer; and

a second oxide or oxynitride layer on the tensile stressed silicon nitride layer,

wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount from between about 8 wt. % and about 12 wt. % at the silicon interface to between about 30 wt. % and about 100 wt. % in a direction away from the P-channel and N-channel transistors.

REMARKS

Claims 14, 16 through 19, and 21 through 35 are pending in this Application. Claims 1 through 13, 15, and 20 were previously canceled. Claims 14, 25, and 35 are currently amended. Care has been exercised to avoid the introduction of new matter. Adequate descriptive support for the present Amendment should be apparent throughout the originally filed disclosure as, for example, in paragraph [0036] of the written description of the specification. Applicants submit that the present Amendment does not generate any new matter issue.

Telephonic Interview

Applicants express appreciation for Examiner Nicely's courtesy in granting and conducting a telephonic interview on July 24, 2012. During the interview, a proposed amendment which has been implemented herewith and the rejections under 35 U.S.C. § 102(b) and 103 were discussed. In addition during the interview, the Examiner courteously suggested reciting the platinum content at the silicon interface, which has also been incorporated in this amendment. The Examiner agreed during the interview that the proposed amendment, clarifying that the platinum composition gradient starts at a nonzero value at the silicon interface and spans the entire depth of the nickel silicide, appears to overcome the rejections relying upon *Okuno* for a platinum gradient. The Examiner, however, indicated that he still needed to update his search.

The specification was objected to as including informalities.

The Examiner stated on page 2 of the Office Action that in paragraph [0006] "compressive strain.." should be "compressive strain." Further, in paragraphs [0042] and

[0043], layer 701 should be a Si-Ge layer and layer 703 should be a Si layer. Last, in paragraph [0049], “e.g.” should be “e.g.”. The specification has been clarified accordingly.

Claims 25 and 35 were objected to as including informalities.

Specifically, on pages 2-3, the Examiner stated that in claim 25 “substrate,” should be “substrate;” and in claim 35, “electrode,” should be “electrode”. Claims 25 and 35 have been clarified accordingly.

Claim 14 is rejected under 35 U.S.C. 102(b) as being anticipated by *Okuno et al.* (U.S. Pub. 2007/0093047).

In the statement of the rejection, the Examiner asserted that *Okuno* discloses a method identically corresponding to that claimed. This rejection is traversed.

The factual determination of lack of novelty under 35 U.S.C. §102(b) requires the identical disclosure in a single reference of each element of a claimed invention, as those elements are set forth in the claims, such that the claimed invention is placed into the recognized possession of one having ordinary skill in the art. *Therasense, Inc. v. Beckton, Dickenson and Company*, 593 F.3d 1289 (Fed. Cir. 2010); *Praxair, Inc. v. ATMI, Inc.*, 543 F.3d 1308, (Fed. Cir. 2008); *Dayco Prods., Inc. v. Total Containment, Inc.*, 329 F.3d 1358 (Fed. Cir. 2003); *Crown Operations International Ltd. v. Solutia Inc.*, 289 F.3d 1367 (Fed. Cir. 2002); *Candt Tech Ltd. v. Resco Metal & Plastics Corp.*, 264 F.3d 1344 (Fed. Cir. 2001). Further, as a matter of procedural due process of law, the Examiner is required to specifically identify where in an applied reference is alleged to disclose each and every feature of a claimed invention, particularly when such is not apparent as in the present case. *In re Rijckaert*, 9 F.3d 1531 (Fed. Cir. 1993);

Lindemann Maschinenfabrik GMBH v. American Hoist & Derrick Co., 730 F.2d 1452 (Fed. Cir. 1984). Indeed, there are fundamental differences between the claimed semiconductor method and *Okuno*'s method that scotch the factual determination that *Okuno* discloses, or even remotely suggests, a method identically corresponding to that claimed.

Specifically, as courteously suggested by the Examiner during the interview, claims 14, 25, and 35 have been clarified by reciting that the gradient begins with an amount of 8 to 12 % platinum at the silicon surface and spans the entire depth of the NiSi layer. As agreed during the interview, *Okuno* fails to teach or suggest a platinum composition gradient that begins at the silicon surface with a nonzero amount and spans the entire depth of the silicide layer. Paragraphs [0130] and [0147] of *Okuno*, referenced by the Examiner at page 3 of the Office Action, do suggest a nickel silicide including platinum having a composition gradient with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode. However, paragraph [0130] further states that the gradient is formed as a platinum containing NiSi layer at the top surface of the NiSi. Therefore, the gradient does not begin at the silicon surface with a nonzero amount and increase across the entire NiSi layer from the silicon surface to the opposite surface. The gradient can only be across a top portion of the NiSi. Further, since the gradient is formed by first forming NiSi and then depositing a very thin layer of platinum over the NiSi and heating (see for example paragraphs [0135]-[0137]), the platinum could not extend to or close to the silicon surface to extend the gradient across the entire NiSi layer, nor is there any teaching or suggestion to do so.

Since *Okuno* does not disclose all of the limitations of independent claim 14 arranged as in the claim, Applicants, therefore, submit that the imposed rejection of claim 14 under 35 U.S.C.

§ 102 (b) for lack of novelty as evidenced by *Okuno* is not factually viable and, hence, solicit withdrawal thereof.

Claims 16 and 17 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Okuno* in view of *Dussarrat et al.* (U.S. Pub. 2010/0256405) in view of *Lee et al.* (“Fully Silicided $\text{Ni}_{1-x}\text{PT}_x\text{Si}$ Metal Gate Electrode for p-MOSFETs) in view of *Cabral et al.* (U.S. Pub. 2008/0274611).

Claims 18 and 19 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Okuno* in view of *Wang et al.* (U.S. Pub. 2005/0095799).

Claim 21 is rejected under 35 U.S.C. 103(a) as being unpatentable over *Okuno* in view of Applicant’s Admitted Prior Art (*AAPA*).

Claims 22 through 24 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Okuno* in view of *Sun et al.* (U.S. Pub. 2005/0247926).

The above-identified rejections are traversed.

Specifically, claims 16 through 19 and 21 through 24 depend ultimately from claim 14. Applicants incorporate herein the arguments previously advanced in traversing the imposed rejection of claim 14 under 35 U.S.C. §102(b) for lack of novelty as evidenced by *Okuno*.

In addition, as agreed during the interview, modifying *Okuno*’s gradient to span the entire depth of the NiSi would be contrary to the explicit teachings of *Okuno* to employ platinum only near the top surface of the NiSi. Accordingly, the skilled artisan would not have been motivated to include a platinum composition gradient in the NiSi starting at a nonzero amount at the silicon surface and spanning the entire depth of the NiSi layer.

Regarding claims 16 and 17, the Office Action (pages 4-6) states that *Dussarratt* teaches a range of 5-10 %, *Lee* discloses a range of 5-33 %, and *Okuno* teaches a gradient, and that the skilled artisan “would have had it within his or her ordinary capabilities through routine experimentation and optimization to arrive at platinum concentrations at the lower and upper ends of 505 that are within the respective claimed ranges as applicant has not disclosed that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical.” However, Applicants explain in paragraph [0037] of the specification that the particular gradient shields NiSi during etching or middle-of-the-line (MOL) processing without undesirably increasing the resistivity of the NiSi. Therefore, the ranges are critical.

Further, although the references may suggest that the platinum concentration affects the electrical resistance and heat-resistant temperature of the nickel silicide, the teachings apply to a uniform nickel platinum silicide, with no indication as to how forming a gradient of platinum concentration would affect the NiSi characteristics. For example, if a higher platinum percentage yields a better work function, the references would suggest employing a high percentage of platinum across the entire silicide layer, not merely at one surface. In addition, as agreed during the interview, since none of *Dussarratt*, *Lee*, and *Cabral* teaches or suggests a composition gradient for the platinum, they cannot, alone or in combination, teach or suggest a platinum composition gradient beginning with a nonzero value at the silicon surface and spanning the entire depth of the silicide layer, and, therefore, do not overcome the deficiencies of *Okuno*.

Similarly, *Wang*, *AAPA*, and *Sun* do not teach or suggest including platinum in the NiSi layer, no less including a composition gradient of platinum. Therefore, *Wang*, *AAPA*, and *Sun*, alone or in combination, do not cure the shortcomings of *Okuno*. Accordingly, even if *Okuno*’s

method were modified as proposed by the Examiner, and Applicants certainly do not agree that the requisite basis to support the asserted motivation has been established, the claimed invention would not result. *Uniroyal, Inc. v. Rudkin-Wiley Corp.*, 837 F.2d 1044 (Fed. Cir. 1988). Applicants therefore submit that the imposed rejections of claims 16 through 19 and 21 through 24, rejected under 35 U.S.C. §103(a) for obviousness predicated upon *Okuno* in view of *Wang*, *AAPA*, or *Sun*, are not factually or legally viable and, hence, solicit withdrawal thereof.

Claims 14, 22 through 27, 33, and 34 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Sun* in view of *Okuno*.

Claims 16, 17, 28, and 29 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Sun* in view of *Okuno* as applied to claims 14 and 27 above, and further in view of *Dussarrat* in view of *Lee* in view of *Cabral*.

Claims 18, 19, 30, and 31 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Sun* in view of *Okuno* as applied to claims 14 and 27 above, and further in view of *Wang*.

Claims 21 and 32 are rejected under 35 U.S.C. 103(a) as being unpatentable over *Sun* in view of *Okuno* as applied to claims 14 and 27 above, and further in view of *AAPA*.

Claim 35 is rejected under 35 U.S.C. 103(a) as being unpatentable over *Sun* in view of *Okuno*, *AAPA*, *Wang*, *Dussarrat*, *Lee*, and *Cabral*.

The above-identified rejections are traversed.

Sun discloses a NiSi layer with no teaching or suggestion to include platinum or a platinum gradient. The Office Action (at pages 9-10) states that it would have been obvious to include a platinum concentration gradient in the NiSi of *Sun* as taught by *Okuno*. However, as

agreed during the interview, *Okuno* does not teach or suggest a platinum gradient across the entire NiSi layer beginning with a nonzero amount at the silicon surface. Further, modifying *Okuno*'s gradient to span the entire depth of the NiSi would be contrary to the explicit teachings of *Okuno* to employ platinum only near the top surface of the NiSi. Accordingly, the skilled artisan would not have been motivated to include a platinum composition gradient in *Sun*'s NiSi layer starting at a nonzero amount at the silicon surface and spanning the entire depth of the NiSi layer. In other words, even if *Sun*'s method were modified in view of *Okuno*, as proposed by the Examiner, and Applicants certainly do not agree that the requisite basis to support the asserted motivation has been established, the claimed invention would not result. See *Uniroyal*.

In addition, regarding claims 16, 17, 28, and 29, the Office Action (pages 11-13) states that *Dussarratt* teaches a range of 5-10 %, *Lee* discloses a range of 5-33 %, and *Okuno* teaches a gradient, and that the skilled artisan "would have had it within his or her ordinary capabilities through routine experimentation and optimization to arrive at platinum concentrations at the lower and upper ends of 505 that are within the respective claimed ranges as applicant has not disclosed that the limitations are for a particular unobvious purpose, produce an unexpected result, or are otherwise critical." However, as discussed *supra*, Applicants explain in paragraph [0037] of the specification that the particular gradient shields NiSi during etching or middle-of-the-line (MOL) processing without undesirably increasing the resistivity of the NiSi. Therefore, the ranges are critical.

Further, although the references may suggest that the platinum concentration affects the electrical resistance and heat-resistant temperature of the nickel silicide, the teachings apply to a uniform nickel platinum silicide, with no indication as to how forming a gradient of platinum concentration would affect the NiSi characteristics. For example, if a higher platinum

percentage yields a better work function, the references would suggest employing a high percentage of platinum across the entire silicide layer, not merely at one surface. In addition, as agreed during the interview, since none of *Dussarratt*, *Lee*, and *Cabral* teaches or suggests a composition gradient for the platinum, they cannot, alone or in combination, teach or suggest a platinum composition gradient beginning with a nonzero value at the silicon surface and spanning the entire depth of the silicide layer, and, therefore, do not overcome the deficiencies of *Sun* in view of *Okuno*.

Furthermore, *AAPA* and *Wang* do not teach or suggest a platinum gradient across a NiSi layer. Therefore, *AAPA*, *Wang*, *Dussarrat*, *Lee*, and *Cabral*, alone or in combination, cannot cure the deficiencies of *Sun* and *Okuno*. Accordingly, even if *Sun*'s method were modified in view of *Okuno* and *AAPA*, *Wang*, *Dussarrat*, *Lee*, and/or *Cabral*, as proposed by the Examiner, and Applicants certainly do not agree that the requisite basis to support the asserted motivation has been established, the claimed invention would not result. See *Uniroyal*. Applicants therefore submit that the imposed rejections of claims 14, 16 through 19 and 21 through 35, rejected under 35 U.S.C. §103(a) for obviousness predicated upon *Sun* in view of *Okuno*, *AAPA*, *Wang*, *Dussarrat*, *Lee*, and/or *Cabral*, are not factually or legally viable and, hence, solicit withdrawal thereof.

Therefore, the present application, as amended, overcomes the objections and rejections of record and is in condition for allowance. Favorable consideration is respectfully requested. If any unresolved issues remain, it is respectfully requested that the Examiner telephone the undersigned attorney at (703) 519-9954 so that such issues may be resolved as expeditiously as possible.

As Applicants' remarks with respect to the Examiner's rejections are sufficient to overcome these rejections, Applicants' silence as to assertions by the Examiner in the Office Action or certain requirements that may be applicable to such rejections (e.g., whether a reference constitutes prior art, ability to combine references, assertions as to patentability of dependent claims) is not a concession by Applicants that such assertions are accurate or such requirements have been met, and Applicants reserve the right to analyze and dispute such assertions in the future.

To the extent necessary, a petition for an extension of time under 37 C.F.R. § 1.136 is hereby made. Please charge any shortage in fees due in connection with the filing of this paper, including extension of time fees, to Deposit Account 504213 and please credit any excess fees to such deposit account.

Respectfully Submitted,

DITTHAVONG MORI & STEINER, P.C.

July 26, 2012
Date

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Electronic Acknowledgement Receipt

EFS ID:	13344451
Application Number:	13096511
International Application Number:	
Confirmation Number:	4709
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NiSi
First Named Inventor/Applicant Name:	Karthik Ramani
Customer Number:	99963
Filer:	Arthur J. Steiner/Vilavanh Ditthavong
Filer Authorized By:	Arthur J. Steiner
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If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 13/096,511		Filing Date 04/28/2011		☐ To be Mailed				
APPLICATION AS FILED – PART I													
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY					
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)					
☒ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A				N/A			
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))		N/A		N/A		N/A				N/A			
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A				N/A			
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 =		*		X \$ =		OR		X \$ =			
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 =		*		X \$ =		OR		X \$ =			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).											
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))													
* If the difference in column 1 is less than zero, enter "0" in column 2.													
APPLICATION AS AMENDED – PART II													
(Column 1)			(Column 2)			(Column 3)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	07/26/2012		CLAIMS REMAINING AFTER AMENDMENT				HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA				
	Total (37 CFR 1.16(i))		* 20		Minus		** 20		= 0				
	Independent (37 CFR 1.16(h))		* 3		Minus		*** 3		= 0				
	☐ Application Size Fee (37 CFR 1.16(s))												
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))												
						TOTAL ADD'L FEE		OR		TOTAL ADD'L FEE			
								OR		0			
(Column 1)			(Column 2)			(Column 3)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT			CLAIMS REMAINING AFTER AMENDMENT				HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA				
	Total (37 CFR 1.16(i))		*		Minus		**		=				
	Independent (37 CFR 1.16(h))		*		Minus		***		=				
	☐ Application Size Fee (37 CFR 1.16(s))												
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))												
						TOTAL ADD'L FEE		OR		TOTAL ADD'L FEE			
								OR					

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.
 ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".
 *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".
 The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

Legal Instrument Examiner:
/GWENDOLYN MYERS/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/096,511	04/28/2011	Karthik Ramani	P2343US01	4709
99963 7590 08/02/2012 DITTHAVONG MORI & STEINER, P.C. 918 PRINCE STREET ALEXANDRIA, VA 22314			EXAMINER NICELY, JOSEPH C	
			ART UNIT 2813	PAPER NUMBER
			NOTIFICATION DATE 08/02/2012	DELIVERY MODE ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

doctet@dcpatent.com

Applicant-Initiated Interview Summary	Application No. 13/096,511	Applicant(s) RAMANI ET AL.	
	Examiner Joseph C. Nicely	Art Unit 2813	

All participants (applicant, applicant's representative, PTO personnel):

(1) Joseph C. Nicely. (3) ____.

(2) Anita Gross (Reg. No. 63,325). (4) ____.

Date of Interview: 24 July 2012.

Type: ☒ Telephonic ☐ Video Conference
 ☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
 If Yes, brief description: _____.

Issues Discussed ☐101 ☐112 ☒102 ☒103 ☐Others
 (For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 14,25 and 35.

Identification of prior art discussed: Okuno et al (US 2007/0093047 and Okuno hereinafter).

Substance of Interview
 (For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Applicant's representative contacted the Examiner to discuss the Okuno prior art document as well as proposed amendments to the claims. Examiner indicated that the proposed amendments to overcome the art needed to include not only that there is a platinum concentration gradient from the silicon interface with the silicide to the opposite surface of the silicide, the platinum concentration gradient at the silicon interface should be amended into the independent claims since the term gradient can include a zero to non-zero transition because gradient merely means a change in value of a quantity (platinum concentration) over a variable (depth within gate electrode/nickel silicide). Since the variable has no defined spacing or increment and change has no defined increment, without introducing that the silicon interface has a non-zero platinum concentration of 8 wt % to 12 wt %, the art could still read on the proposed amendments. However, the inclusion of the 8 wt % to 12 wt % at the silicon interface into the independent claims would overcome Okuno..

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☐ Attachment

/J. C. N./ Examiner, Art Unit 2813	/Matthew C. Landau/ Supervisory Patent Examiner, Art Unit 2813
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Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews

Paragraph (b)

In every instance where reconsideration is requested in view of an interview with an examiner, a complete written statement of the reasons presented at the interview as warranting favorable action must be filed by the applicant. An interview does not remove the necessity for reply to Office action as specified in §§ 1.111, 1.135. (35 U.S.C. 132)

37 CFR §1.2 Business to be transacted in writing.

All business with the Patent or Trademark Office should be transacted in writing. The personal attendance of applicants or their attorneys or agents at the Patent and Trademark Office is unnecessary. The action of the Patent and Trademark Office will be based exclusively on the written record in the Office. No attention will be paid to any alleged oral promise, stipulation, or understanding in relation to which there is disagreement or doubt.

The action of the Patent and Trademark Office cannot be based exclusively on the written record in the Office if that record is itself incomplete through the failure to record the substance of interviews.

It is the responsibility of the applicant or the attorney or agent to make the substance of an interview of record in the application file, unless the examiner indicates he or she will do so. It is the examiner's responsibility to see that such a record is made and to correct material inaccuracies which bear directly on the question of patentability.

Examiners must complete an Interview Summary Form for each interview held where a matter of substance has been discussed during the interview by checking the appropriate boxes and filling in the blanks. Discussions regarding only procedural matters, directed solely to restriction requirements for which interview recordation is otherwise provided for in Section 812.01 of the Manual of Patent Examining Procedure, or pointing out typographical errors or unreadable script in Office actions or the like, are excluded from the interview recordation procedures below. Where the substance of an interview is completely recorded in an Examiners Amendment, no separate Interview Summary Record is required.

The Interview Summary Form shall be given an appropriate Paper No., placed in the right hand portion of the file, and listed on the "Contents" section of the file wrapper. In a personal interview, a duplicate of the Form is given to the applicant (or attorney or agent) at the conclusion of the interview. In the case of a telephone or video-conference interview, the copy is mailed to the applicant's correspondence address either with or prior to the next official communication. If additional correspondence from the examiner is not likely before an allowance or if other circumstances dictate, the Form should be mailed promptly after the interview rather than with the next official communication.

The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
- The signature of the examiner who conducted the interview (if Form is not an attachment to a signed Office action)

It is desirable that the examiner orally remind the applicant of his or her obligation to record the substance of the interview of each case. It should be noted, however, that the Interview Summary Form will not normally be considered a complete and proper recordation of the interview unless it includes, or is supplemented by the applicant or the examiner to include, all of the applicable items required below concerning the substance of the interview.

A complete and proper recordation of the substance of any interview should include at least the following applicable items:

- 1) A brief description of the nature of any exhibit shown or any demonstration conducted,
- 2) an identification of the claims discussed,
- 3) an identification of the specific prior art discussed,
- 4) an identification of the principal proposed amendments of a substantive nature discussed, unless these are already described on the Interview Summary Form completed by the Examiner,
- 5) a brief identification of the general thrust of the principal arguments presented to the examiner,
(The identification of arguments need not be lengthy or elaborate. A verbatim or highly detailed description of the arguments is not required. The identification of the arguments is sufficient if the general nature or thrust of the principal arguments made to the examiner can be understood in the context of the application file. Of course, the applicant may desire to emphasize and fully describe those arguments which he or she feels were or might be persuasive to the examiner.)
- 6) a general indication of any other pertinent matters discussed, and
- 7) if appropriate, the general results or outcome of the interview unless already described in the Interview Summary Form completed by the examiner.

Examiners are expected to carefully review the applicant's record of the substance of an interview. If the record is not complete and accurate, the examiner will give the applicant an extendable one month time period to correct the record.

Examiner to Check for Accuracy

If the claims are allowable for other reasons of record, the examiner should send a letter setting forth the examiner's version of the statement attributed to him or her. If the record is complete and accurate, the examiner should place the indication, "Interview Record OK" on the paper recording the substance of the interview along with the date and the examiner's initials.

To: docket@dcpatent.com,,
From: PAIR_eOfficeAction@uspto.gov
Cc: PAIR_eOfficeAction@uspto.gov
Subject: Private PAIR Correspondence Notification for Customer Number 99963

Aug 02, 2012 05:25:07 AM

Dear PAIR Customer:

DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314
UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 99963 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

Disclaimer:

The list of documents shown below is provided as a courtesy and is not part of the official file wrapper. The content of the images shown in PAIR is the official record.

Application	Document	Mailroom Date	Attorney Docket No.
13096511	INTV.SUM.APP	08/02/2012	P2343US01

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>.

If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov with 'e-Office Action' on the subject line or call 1-866-217-9197 during the following hours:

Monday - Friday 6:00 a.m. to 12:00 a.m.

Thank you for prompt attention to this notice,

UNITED STATES PATENT AND TRADEMARK OFFICE
PATENT APPLICATION INFORMATION RETRIEVAL SYSTEM



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NOTICE OF ALLOWANCE AND FEE(S) DUE

99963 7590 08/15/2012
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

EXAMINER

NICELY, JOSEPH C

ART UNIT

PAPER NUMBER

2813

DATE MAILED: 08/15/2012

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/096,511

04/28/2011

Karthik Ramani

P2343US01

4709

TITLE OF INVENTION: METHOD TO REDUCE MOL DAMAGE ON NISI

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1740	\$300	\$0	\$2040	11/15/2012

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the SMALL ENTITY status shown above.

If the SMALL ENTITY is shown as YES, verify your current SMALL ENTITY status:

A. If the status is the same, pay the TOTAL FEE(S) DUE shown above.

B. If the status above is to be removed, check box 5b on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and twice the amount of the ISSUE FEE shown above, or

If the SMALL ENTITY is shown as NO:

A. Pay TOTAL FEE(S) DUE shown above, or

B. If applicant claimed SMALL ENTITY status before, or is now claiming SMALL ENTITY status, check box 5a on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and 1/2 the ISSUE FEE shown above.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax **(571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

99963 7590 08/15/2012
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/096,511 04/28/2011 Karthik Ramani P2343US01 4709

TITLE OF INVENTION: METHOD TO REDUCE MOL DAMAGE ON NISI

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
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nonprovisional NO \$1740 \$300 \$0 \$2040 11/15/2012

EXAMINER	ART UNIT	CLASS-SUBCLASS
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NICELY, JOSEPH C 2813 257-413000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
- (2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
- 3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
- ☐ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
- ☐ Payment by credit card. Form PTO-2038 is attached.
- ☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/096,511	04/28/2011	Karthik Ramani	P2343US01	4709
99963 7590 08/15/2012 DITTHAVONG MORI & STEINER, P.C. 918 PRINCE STREET ALEXANDRIA, VA 22314			EXAMINER NICELY, JOSEPH C	
			ART UNIT 2813	PAPER NUMBER
DATE MAILED: 08/15/2012				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment to date is 0 day(s). If the issue fee is paid on the date that is three months after the mailing date of this notice and the patent issues on the Tuesday before the date that is 28 weeks (six and a half months) after the mailing date of this notice, the Patent Term Adjustment will be 0 day(s).

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No.	Applicant(s)	
	13/096,511	RAMANI ET AL.	
	Examiner	Art Unit	
	Joseph C. Nicely	2813	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the amendment filed 07/26/2012.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 14 and 16-35.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
 - a) ☐ All b) ☐ Some* c) ☐ None of the:
 1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ A SUBSTITUTE OATH OR DECLARATION must be submitted. Note the attached EXAMINER'S AMENDMENT or NOTICE OF INFORMAL PATENT APPLICATION (PTO-152) which gives reason(s) why the oath or declaration is deficient.
6. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
 - (a) ☐ including changes required by the Notice of Draftsperson's Patent Drawing Review (PTO-948) attached
 - 1) ☐ hereto or 2) ☐ to Paper No./Mail Date ____.
 - (b) ☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

7. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|--|
| <ol style="list-style-type: none"> 1. ☒ Notice of References Cited (PTO-892) 2. ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) 3. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date ____ 4. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | <ol style="list-style-type: none"> 5. ☐ Notice of Informal Patent Application 6. ☐ Interview Summary (PTO-413),
Paper No./Mail Date ____. 7. ☐ Examiner's Amendment/Comment 8. ☒ Examiner's Statement of Reasons for Allowance 9. ☐ Other ____. |
|---|--|

/J. C. N./
Examiner, Art Unit 2813

/Matthew C. Landau/
Supervisory Patent Examiner, Art Unit 2813

DETAILED ACTION

1. This Office action is in response to the amendment filed 07/26/2012 in which claims 14, 25, and 35 were amended.
2. Claims 14, 16-19, and 21-35 remain pending.

REASONS FOR ALLOWANCE

3. Claims 14, 16-19, and 21-35 are allowed over the prior art of record.
4. The following is an examiner's statement of reasons for allowance:

As to claims 14, 25, and 35: the prior art of record fails to teach in combination a semiconductor device where the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode (the P-channel and N-channel transistors), from between about 8 wt. % and about 12 wt. % at the silicon interface, and including all limitations.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

5. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.
 - a. Okuno et al. (US 2007/0093047) discloses that the platinum in the nickel silicide exists only at the top portion of the nickel silicide, thus, there is no

concentration of platinum at the silicon interface from between about 8 wt. % and about 12 wt. % and that increases in concentration a direction away from the gate electrode/(P or N) channel transistors.

b. Tsai et al (US 7,768,072) discloses fully silicided gates with a platinum concentration that is greatest near the gate dielectric, thus, the platinum concentration does not increase in a direction away from the source/drain regions/(P or N) channel transistors.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Joseph C. Nicely whose telephone number is (571)270-3834. The examiner can normally be reached on Monday through Friday 8:30AM-6:00PM EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Matthew Landau can be reached on (571) 272-1731. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2813

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/J. C. N./
Examiner, Art Unit 2813

/Matthew C. Landau/
Supervisory Patent Examiner, Art Unit 2813

Notice of References Cited	Application/Control No. 13/096,511	Applicant(s)/Patent Under Reexamination RAMANI ET AL.	
	Examiner Joseph C. Nicely	Art Unit 2813	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-7,768,072 B2	08-2010	Tsai et al.	257/368
	B	US-			
	C	US-			
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.



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 www.uspto.gov

BIB DATA SHEET

CONFIRMATION NO. 4709

SERIAL NUMBER 13/096,511	FILING or 371(c) DATE 04/28/2011 RULE	CLASS 257	GROUP ART UNIT 2813	ATTORNEY DOCKET NO. P2343US01		
APPLICANTS Karthik Ramani, Hopewell Junction, NY; Paul R. BESSER, Sunnyvale, CA; ** CONTINUING DATA ***** This application is a DIV of 12/366,378 02/05/2009 PAT 7,994,038 ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 05/10/2011						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /JOSEPH C NICELY/ Acknowledged Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY NY	SHEETS DRAWINGS 13	TOTAL CLAIMS 20	INDEPENDENT CLAIMS 3
ADDRESS DITTHAVONG MORI & STEINER, P.C. 918 PRINCE STREET ALEXANDRIA, VA 22314 UNITED STATES						
TITLE METHOD TO REDUCE MOL DAMAGE ON NiSi						
FILING FEE RECEIVED 1090	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees		
				☐ 1.16 Fees (Filing)		
				☐ 1.17 Fees (Processing Ext. of time)		
				☐ 1.18 Fees (Issue)		
				☐ Other _____		
				☐ Credit		

<i>Index of Claims</i> 	Application/Control No. 13096511	Applicant(s)/Patent Under Reexamination RAMANI ET AL.
	Examiner JOSEPH C NICELY	Art Unit 2813

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant				☐ CPA				☐ T.D.				☐ R.1.47			
CLAIM		DATE													
Final	Original	04/11/2012	08/01/2012												
	1	-	-												
	2	-	-												
	3	-	-												
	4	-	-												
	5	-	-												
	6	-	-												
	7	-	-												
	8	-	-												
	9	-	-												
	10	-	-												
	11	-	-												
	12	-	-												
	13	-	-												
1	14	✓	=												
	15	-	-												
2	16	✓	=												
3	17	✓	=												
4	18	✓	=												
5	19	✓	=												
	20	-	-												
6	21	✓	=												
7	22	✓	=												
8	23	✓	=												
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10	25	✓	=												
11	26	✓	=												
12	27	✓	=												
13	28	✓	=												
14	29	✓	=												
15	30	✓	=												
16	31	✓	=												
17	32	✓	=												
18	33	✓	=												
19	34	✓	=												
20	35	✓	=												

Search Notes 	Application/Control No. 13096511	Applicant(s)/Patent Under Reexamination RAMANI ET AL.
	Examiner JOSEPH C NICELY	Art Unit 2813

SEARCHED			
Class	Subclass	Date	Examiner
257	369, 383, 412, E29.139, E29.242	4/11/2012	JCN
257	382, 383, 388, 412, 413, E29.156, 377	7/31/2012	JCN

SEARCH NOTES		
Search Notes	Date	Examiner
EAST Search (see attached history)	4/11/2012	JCN
Google Search (see attached history)	4/11/2012	JCN
IEEE Xplore Search (see attached history)	4/11/2012	JCN
Inventor Name Search and Assignee Search conducted through EAST	4/11/2012	JCN
EAST Search (see attached history)	7/31/2012	JCN

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner
257	413, 383, E29.156	7/31/2012	JCN

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	11	("20050095799" "20070020929" "20050247926" "20090028744" "20070093047" "20080237750" "20100256405" "20080274611" "6841449" "20070034967" "20080090369").PN.	US-PGPUB; USPAT; FPRS; EPO; JPO; IBM_TDB	ADJ	ON	2012/07/31 16:17
S2	5	("0689688" "20070138580" "20070090417" "20050045965" "20050110070").PN.	US-PGPUB; USPAT; FPRS; EPO; JPO; IBM_TDB	ADJ	ON	2012/07/31 16:39
S3	2	("2007/0093047").URPN.	USPAT	ADJ	ON	2012/07/31 16:46
S4	163	(silicide OR nickel near2 silicide OR NiSi) WITH (platinum OR Pt) WITH (concentration OR gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 16:48
S5	1828	257/412.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:12
S6	397	S5 and (Pt OR platinum)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:13
S7	196	S5 and (Pt OR platinum) WITH (silicide)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:13
S8	190	S7 not S4	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:13
S9	5	S8 AND (platinum OR Pt) WITH (concentration OR gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:13
S10	735	257/413.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:14
S11	94	S10 and (Pt OR platinum) WITH (silicide)	US-PGPUB; USPAT; FPRS;	ADJ	ON	2012/07/31 17:14

			EPO; JPO; DERWENT; IBM_TDB			
S12	90	S11 noT (S4 or S9)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:14
S13	0	S12 AND (platinum OR Pt) WITH (concentration OR gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:14
S14	340	257/388.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:15
S15	4	S14 AND (platinum OR Pt) WITH (concentration OR gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:15
S16	827	257/E29.156.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:16
S17	2	S16 AND (platinum OR Pt) WITH (concentration OR gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:17
S18	468	257/377.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:26
S19	5	S18 AND (platinum OR Pt) WITH (concentration OR gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:26
S20	1341	257/382,383.ccls.	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:26
S21	10	S20 AND (platinum OR Pt) WITH (concentration OR gradient)	US-PGPUB; USPAT; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/07/31 17:26

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S22	25	(silicide OR nickel near2 silicide OR NiSi).clm. WITH (platinum OR Pt).clm.	US-PGPUB;	ADJ	ON	2012/07/31 17:28

		WITH (concentration OR gradient).clm.	USPAT; UPAD			
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8/ 1/ 2012 11:48:14 AM
C:\ Users\ jnicely\ Documents\ EAST\ Workspaces\ Allowable\ 13096511.wsp

To: docket@dcpatent.com,,
From: PAIR_eOfficeAction@uspto.gov
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Aug 15, 2012 05:23:26 AM

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ALEXANDRIA, VA 22314
UNITED STATES

The following USPTO patent application(s) associated with your Customer Number, 99963 , have new outgoing correspondence. This correspondence is now available for viewing in Private PAIR.

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Application	Document	Mailroom Date	Attorney Docket No.
13096511	NOA	08/15/2012	P2343US01
	NOA	08/15/2012	P2343US01
	892	08/15/2012	P2343US01

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Alexandria, Virginia 22313-1450
or Fax **(571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

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99963 7590 08/15/2012
DITTHAVONG MORI & STEINER, P.C.
918 PRINCE STREET
ALEXANDRIA, VA 22314

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

13/096,511 04/28/2011 Karthik Ramani P2343US01 4709

TITLE OF INVENTION: METHOD TO REDUCE MOL DAMAGE ON NISI

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
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nonprovisional NO \$1740 \$300 \$0 \$2040 11/15/2012

EXAMINER	ART UNIT	CLASS-SUBCLASS
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NICELY, JOSEPH C 2813 257-413000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) the names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Ditthavong Mori & Steiner, P.C.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

GLOBALFOUNDRIES INC.

Grand Cayman, Cayman Islands

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

☒ Issue Fee

☐ Publication Fee (No small entity discount permitted)

☐ Advance Order - # of Copies _____

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☐ A check is enclosed.

☒ Payment by credit card. Form PTO-2038 is attached.

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number 504213 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27.

☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature /Phouphanomketh Ditthavong/

Date November 8, 2012

Typed or printed name Phouphanomketh Ditthavong

Registration No. 44658

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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Electronic Patent Application Fee Transmittal				
Application Number:		13096511		
Filing Date:		28-Apr-2011		
Title of Invention:		METHOD TO REDUCE MOL DAMAGE ON NISI		
First Named Inventor/Applicant Name:		Karthik Ramani		
Filer:		Phouphanomketh Ditthavong/Tiffany Butler		
Attorney Docket Number:		P2343US01		
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl issue fee	1501	1	1770	1770
Publ. Fee- early, voluntary, or normal	1504	1	300	300

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				2070

Electronic Acknowledgement Receipt

EFS ID:	14185075
Application Number:	13096511
International Application Number:	
Confirmation Number:	4709
Title of Invention:	METHOD TO REDUCE MOL DAMAGE ON NISI
First Named Inventor/Applicant Name:	Karthik Ramani
Customer Number:	99963
Filer:	Phouphanomketh Ditthavong/Tiffany Butler
Filer Authorized By:	Phouphanomketh Ditthavong
Attorney Docket Number:	P2343US01
Receipt Date:	08-NOV-2012
Filing Date:	28-APR-2011
Time Stamp:	17:27:51
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 2070
RAM confirmation Number	4867
Deposit Account	504213
Authorized User	DITTHAVONG, PHOUPHANOMKETH

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Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

GF-ITC-00004481

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Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	NY0281D1_P2343US01_IssueFeeTransmittal_filed11-08-2012.pdf	183339 e6d23cedb818637b5c2f2f86588e533d42e70ec3	no	1
Warnings:					
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2	Fee Worksheet (SB06)	fee-info.pdf	32026 994e24c756174f23f3d6207f5a90d6c2d86fe83c	no	2
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13/096,511	12/11/2012	8330235	P2343US01	4709

99963 7590 11/21/2012
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ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

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APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

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